

Modal Estimation and Inference of Autoregressive Distributed Lag Model Applied to COVID-19 Pandemic Data

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Abstract

This study applies modal regression to the autoregressive distributed lag (ARDL) model and estimates the long- and short-run relationships between integrated series with outliers. Using the proposed modal ARDL (MARDL) methodology, we provide asymptotic distributions of the parameter estimators in the model. We corroborate the theoretical results by simulation of the short- and long-run parameter estimators and Wald tests defined by these estimators. We then apply the MARDL methodology to analyze the short- and long-run relationships between COVID-19 confirmed cases and deaths that are notably characterized by outliers. To assess the dynamic changes in confirmed cases and deaths during the pre- and post-vaccination periods and to examine how the case fatality rate moves together with government response, we examine global data and data from Belgium, Germany, Denmark, and Sweden. Our empirical findings demonstrate that the in-sample forecasts obtained from MARDL broadly outperform those obtained from ARDL and QARDL in terms of the coverage quantile function. Therefore, the MARDL methodology is useful in investigating how the pandemic evolved in response to vaccination and government mitigation efforts.

Key Words: MARDL; Cointegration; Modal Regression; Conditional Mode; COVID-19; Vaccination; Government Response.

JEL Classifications: C22, C53, I18.

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1 Introduction

Cointegration analysis plays an important role in analyzing interrelationships between integrated series. For example, in their classic paper, [Engle and Granger \(1987\)](#) estimated a cointegrating relationship by least squares in a two-step estimation procedure. Another classic paper, [Pesaran and Shin \(1998\)](#), estimated an autoregressive distributed lag (ARDL) model that can be converted to the error-correction (EC) representation to apply the two-step procedure in [Engle and Granger \(1987\)](#).

Conducting estimations and inferences for integrated series is crucial for various purposes. The long-run relationship between integrated series is different from the short-run dynamics between them; for example, a government policy implied by a long-run cointegrating relationship must be framed differently from that implied by short-run dynamics. The more precise the model parameter estimator, the larger the attainable efficacy of government policy. Nevertheless, as [Nielsen \(2004\)](#) and others have pointed out, estimating and inferring unknown parameters are seriously affected by outliers. Standard least squares (SLS) estimation cannot produce short- and long-run parameter estimators invulnerable to outliers.

Our main goal is to develop an estimation and inference method that researchers can trust so that short- and long-run relationships can be estimated without the outliers ruining them. Accordingly, we employ the popular modal regression (MR) method for estimating the conditional mode of stationary data, and call this approach the modal autoregressive distributed lag (MARDL) methodology. We apply this methodology to the data on COVID-19 confirmed cases and deaths to empirically illustrate our method.

The literature records attempts to estimate a cointegrating relationship. For example, [Cho et al. \(2015\)](#) provide the quantile autoregressive distributed lag (QARDL) methodology to estimate short- and long-run relationships by quantile regression, so that parameter estimates depend on percentiles. [Nielsen \(2004\)](#) and [Lyhagen and Baris-Tüemen \(2025\)](#) discover that additive outliers seriously affect parameter estimation by simulation, and [Berenguer-Rico and Nielsen \(2026\)](#) examine least trimmed squares estimation as a resolution. The current study provides another methodology to overcome the challenges raised by outliers by employing MR for cointegration analysis.

Applying MR for cointegration analysis is challenging (details in Section 2). Because regressors are formed by both integrated and stationary series in the context of ARDL, their impact on MR parameter estimation has to be analyzed with care. Instead of applying MR analysis in parallel to MR applied to stationary variables, we separately analyze the long- and short-run parameter estimators by focusing on their roles in the MARDL context. We deliberately conduct this separate analysis because their different roles can be used to define the estimators for the parameters in the EC representation. Using these estimators, we

further define test statistics for the parameters by the Wald test principle. The null limit distributions of the Wald tests are chi-squared, allowing straightforward empirical applications.

MR analysis is widely used for stationary data analysis. [Sager and Thisted \(1982\)](#) first proposed the idea of estimating the conditional mode in isotonic regression, followed by [Collomb et al. \(1986\)](#) who theoretically proved that the non-parametric mode estimator is consistent for the conditional mode functions under certain regularity conditions. Subsequently, [Lee \(1989\)](#) pioneered MR from the perspective of supervised learning and risk minimization by providing a consistent mean estimator for the conditional mode using non-truncated observations and further established a theoretical foundation for MR. [Lee \(1989, 1993\)](#) ensured the consistency of the estimator by requiring that the model has a constant adjustment parameter and an error density function symmetric around zero, which make the proposed MR estimator robust. [Yao et al. \(2012\)](#) demonstrated the consistency of a non-parametric MR estimator under symmetric noise, and [Kemp and Silva \(2012\)](#) showed the consistency and asymptotic normality of a linear MR estimator under an asymmetric density function condition, when the mode is globally unique. [Yao and Li \(2014\)](#) proposed a kernel-based objective function with the bandwidth approaching zero to achieve a consistent mode estimator for skewed data; their approach is popular in the literature (see [Chen et al., 2016](#); [Yao and Xiang, 2016](#); [Krief, 2017](#); [Chen, 2018](#); [Ota et al., 2019](#); [Zhou and Huang, 2019](#); [Yang et al., 2024](#); [Ullah et al., 2022, 2023](#); [Jing and Cho, 2025](#), among others).

We develop the MARDL methodology by applying [Yao and Li's \(2014\)](#) approach to the ARDL framework. We pay particular attention to the convergence rate of the bandwidth in the kernel function for this application. When the unknown parameters are estimated by minimizing the kernel-based objective function, the role of the bandwidth is critical for density smoothing, which affects the convergence rates of the parameter estimators. The convergence rate we specify for parameter estimation allows both short- and long-run parameter estimators to be asymptotically normal, and the smoothing bias is not incurred with their asymptotic distributions. We find that the bias is associated with only the estimation of the intercept, rendering the MARDL methodology straightforward for empirical applications.

We further demonstrate the MARDL methodology by examining the relationship between the data on COVID-19 confirmed cases and deaths that are notably characterized by outliers. We use global data and local data from Belgium, Denmark, Germany, and Sweden to affirm that MARDL broadly achieves coverage quantile function (CQF) levels lower than those achieved under ARDL and QARDL, where CQF measures the median of the absolute in-sample forecast errors. We also demonstrate how to exploit the MARDL methodology in investigating the evolution of the pandemic in response to vaccination and government mitigation efforts.

The remaining paper is structured as follows: in Section 2, we introduce the MARDL methodology that derives the asymptotic distribution of the short- and long-run parameter estimators, and develop testing methodology using the asymptotic distributions. Section 3 evaluates the finite-sample performances of the test statistics in Section 2 by Monte Carlo simulations. In Section 4, we conduct empirical analysis by applying the MARDL methodology to the data of daily confirmed COVID-19 cases and deaths. Finally, Section 5 summarizes and concludes. All mathematical proofs, supplementary simulations, and empirical information are collected in the Online Supplement.

The notations used in this study are as follows: A parameter without a subscript indicates a generic parameter value in the parameter space. If it is attached by a subscript, it characterizes a specific element in the parameter space. We let “ $\rightsquigarrow$,” “ $\xrightarrow{\mathbb{P}}$,” and “ $\lfloor a \rfloor$ ” represent “weak convergence,” “convergence in probability,” and “the largest integer not greater than a ,” respectively. The matrix $[a_{ij}]$ with $i = 1, \dots, m, j = 1, \dots, n$ is an $m \times n$ matrix with the element a_{ij} , and $\mathbf{1}_\ell$ signifies an $\ell \times 1$ vector of ones. We let $D_n(\cdot) = D(\cdot) + o_{\mathbb{P}}(w_n)$ (resp. $O_{\mathbb{P}}(w_n)$) if $\sup_{x \in \mathcal{X}} |D_n(x) - D(x)| = o_{\mathbb{P}}(w_n)$ (resp. $O_{\mathbb{P}}(w_n)$). We express that $f(n) = o(g(n))$ if for all $c > 0$, there exists some $k > 0$ (independent of n) such that $0 \leq f(n) < cg(n)$ for all $n \geq k$. Furthermore, for any real matrix $\mathbf{A}$, we let $\|\mathbf{A}\|$ be the Frobenius norm of $\mathbf{A}$, that is, $\|\mathbf{A}\| = [\text{tr}(\mathbf{A}\mathbf{A}')]^{1/2}$. If $\mathbf{a}$ is a vector, $\|\mathbf{a}\|$ is exactly the same as the Euclidean norm of $\mathbf{a}$. We also abbreviate stochastic integrals by omitting differentials. For example, $\int_0^1 \mathbf{B}_w(r) \mathbf{B}_w(r)' dr$ and $\int_0^1 \mathbf{B}_w(r) d\mathbf{B}_\phi(r)$ are abbreviated as $\int \mathbf{B}_w \mathbf{B}_w'$ and $\int \mathbf{B}_w d\mathbf{B}_\phi$, respectively, where $\mathbf{B}_w(\cdot)$ and $\mathbf{B}_\phi(\cdot)$ are the vectors of Brownian motions. All others are standard.

2 Estimation and Inference of MARDL

Cho et al. (2015) extended the ARDL approach to quantile regression and jointly analyzed the short- and long-run cointegrating relationships across a range of quantiles, now known as QARDL. We extend the QARDL analysis to the context of MR.

The QARDL analysis allows for different cointegrating coefficients depending on percentiles. Specifically, it is defined as

$$y_t = \alpha_*(\tau) + \sum_{j=1}^p \phi_{j*}(\tau) y_{t-j} + \sum_{j=0}^q \theta_{j*}(\tau)' \mathbf{x}_{t-j} + u_t(\tau),$$

commonly referred to as QARDL(p, q), where $\Delta \mathbf{x}_t$ is a k -dimensional stationary ergodic process with zero population mean ($k \in \mathbb{N}$). The error term $u_t(\tau)$ is defined as $y_t - Q_\tau(y_t | \mathcal{F}_{t-1})$, where $Q_\tau(y_t | \mathcal{F}_{t-1})$ is the conditional quantile function on $\mathcal{F}_{t-1}$; $\mathcal{F}_{t-1}$ is the smallest σ -field generated by $\{\mathbf{x}'_t, y_{t-1}, \mathbf{x}'_{t-1}, \dots\}$; and p

and q are the QARDL lag orders.

We define MARDL in parallel to QARDL. If $f(\cdot|\mathcal{F}_{t-1})$ is the conditional probability density function (PDF) of y_t given $\mathcal{F}_{t-1}$, we suppose that $\text{mode}(y_t|\mathcal{F}_{t-1}) := \arg \max_y f(y|\mathcal{F}_{t-1})$ is unique, and that $\text{mode}(y_t|\mathcal{F}_{t-1})$ is given as

$$\text{mode}(y_t|\mathcal{F}_{t-1}) = \alpha_* + \sum_{j=1}^p \phi_{j*} y_{t-j} + \sum_{j=0}^q \theta'_{j*} \mathbf{x}_{t-j},$$

so that y_t can be decomposed as follows:

$$y_t = \alpha_* + \sum_{j=1}^p \phi_{j*} y_{t-j} + \sum_{j=0}^q \theta'_{j*} \mathbf{x}_{t-j} + u_t, \quad (1)$$

which we call MARDL(p, q). Here, $u_t := y_t - \text{mode}(y_t|\mathcal{F}_{t-1})$ is defined as the modal error.

Remark 1. (i) [Nielsen \(2004\)](#) and [Lyhagen and Baris-Tüemen \(2025\)](#) discover that additive outliers seriously affect parameter estimation by simulation when the relationship between integrated series is estimated using SLS. [Berenguer-Rico and Nielsen \(2026\)](#) examine least trimmed squares estimation as a resolution, and the QARDL methodology can also estimate the unknown parameters without being affected by outliers, but it depends on percentiles. This feature motivates us to estimate the unknown parameters by y_t on $\mathcal{F}_{t-1}$.

(ii) The modal error u_t satisfies $\text{mode}(u_t|\mathcal{F}_{t-1}) = 0$ by construction. If the density function of u_t , denoted as $f_u(\cdot|\mathcal{F}_{t-1})$, is smooth, $f_u^{(1)}(0|\mathcal{F}_{t-1}) = 0$ with probability 1. $\square$

2.1 Environment and Parameter Estimation

We now propose an estimation procedure for the long- and short-run parameters in MARDL(p, q).

2.1.1 Error-Correction Representation

A modal cointegrating relationship can be converted into an EC representation in parallel to [Engle and Granger \(1987\)](#). Specifically, we can convert MARDL(p, q) into the following:

$$\Delta y_t = \alpha_* + \zeta_* (y_{t-1} - \beta'_* \mathbf{x}_{t-1}) + \sum_{j=1}^{p-1} \varphi_{j*} \Delta y_{t-j} + \sum_{j=0}^{q-1} \psi'_{j*} \Delta \mathbf{x}_{t-j} + u_t, \quad (2)$$

where the parameters in (2) are defined as

$$\zeta_* := \sum_{i=1}^p \phi_{i*} - 1, \quad \beta_* := \gamma_* \left(1 - \sum_{i=1}^p \phi_{i*} \right)^{-1}, \quad \gamma_* := \sum_{j=0}^q \theta_{j*}, \quad \varphi_{j*} := - \sum_{h=j+1}^p \phi_{h*},$$

$\theta_{j=1}^* := - \sum_{h=j+1}^q \theta_{h*}$ for $j \neq 0$, and $\theta_0^* := \theta_{0*}$. Here, β_* captures the long-run relationship between y_t and $\mathbf{x}_t$, and $\varphi_* := (\varphi_{1*}, \dots, \varphi_{p-1*})'$ and $\psi_* := (\psi'_{0*}, \dots, \psi'_{q-1*})'$ describe the short-run dynamics between the variables.

We envisage the long- and short-run relationships posited by the EC representation in an alternative way. We first rewrite (1) as follows:

$$y_t = \alpha_* + \sum_{j=0}^{q-1} \mathbf{w}'_{t-j} \delta_{j*} + \mathbf{x}'_t \gamma_* + \sum_{j=1}^p \phi_{j*} y_{t-j} + u_t \quad (3)$$

We assume $\mathbf{w}_t := \Delta \mathbf{x}_t$ for notational simplicity, and for each $j = 0, \dots, q-1$, we assume $\delta_{j*} := - \sum_{i=j+1}^q \theta_{i*}$. We further reformulate (3) as

$$y_t = \mu_* + \mathbf{x}'_t \beta_* + r_t, \quad (4)$$

where $\mu_* := \alpha_* (1 - \sum_{i=1}^p \phi_{i*})^{-1}$ and $r_t := \sum_{j=0}^{\infty} \mathbf{w}'_{t-j} \xi_{0,j*} + \sum_{j=0}^{\infty} \rho_{j*} u_{t-j}$, such that $\{\pi_{0*}, \pi_{1*}, \dots\}$ is a set of coefficients satisfying

$$(1 - L)^{-1} \left(\frac{\sum_{j=0}^q \theta_{j*} L^j}{1 - \sum_{j=1}^p \phi_{j*} L^j} - \frac{\sum_{j=0}^q \theta_{j*}}{1 - \sum_{j=1}^p \phi_{j*}} \right) \equiv \sum_{j=0}^{\infty} \pi_{j*} L^j,$$

and for $i = 0, 1, \dots, p$ and $j = 0, 1, \dots$, $\xi_{i,j*} := -\beta_*$, for $j < i$; and $\xi_{i,j*} := \pi_{(j-i)*}$ for $j \geq i$, so that $\xi_{0,j*} = \pi_{j*}$. Likewise, $\{\rho_{0*}, \rho_{1*}, \dots\}$ is a set of coefficients satisfying $\sum_{j=0}^{\infty} \rho_{j*} L^j \equiv (1 - \sum_{j=1}^p \phi_{j*} L^j)^{-1}$. The residual term r_t collects serially correlated stationary variables irrelevant to the long-run relationship, which is captured by β_* , defined by the parameters in the MARDL process, γ_* and $\phi_* := (\phi_{1*}, \dots, \phi_{p*})'$. This means that the long-run parameter β_* is given as

$$\beta_* := \gamma_* \left(1 - \sum_{i=1}^p \phi_{i*} \right)^{-1}. \quad (5)$$

The short-run parameters in (2) can also be represented using the parameters in MARDL(p, q). If we

assume $\delta_* := (\delta'_{0*}, \dots, \delta'_{(q-1)*})'$ and

$$\mathbf{R}_1 := \begin{bmatrix} \mathbf{U} & \boldsymbol{\iota}_{p-1} \\ \mathbf{0}'_{p-1} & 1 \end{bmatrix},$$

such that $\mathbf{U}$ is an upper-triangular matrix with ones for the nonzero elements, then it follows that

$$\boldsymbol{\varphi}_* = -\mathbf{R}_1 \boldsymbol{\phi}_* \quad \text{and} \quad \boldsymbol{\psi}_* = \delta_* + (\boldsymbol{\gamma}'_*, \mathbf{0}'_{k(q-1)})'. \quad (6)$$

2.1.2 Estimation and Environment

To meet the study's goal, we estimate long- and short-run parameters in two steps: first, the long-run parameters by estimating $\boldsymbol{\gamma}_*$ and $\boldsymbol{\phi}_*$, and then, the short-run parameters using the EC representation.

Before estimating the unknown parameters, we rewrite MARDL(p, q) as follows to discuss its notable features:

$$y_t = \mathbf{g}'_t \boldsymbol{\lambda}_* + \widehat{\mathbf{y}}'_t \boldsymbol{\phi}_* + u_t = \mathbf{z}'_t \boldsymbol{\alpha}_* + u_t \quad (7)$$

where

$$\mathbf{z}_t := (\mathbf{g}'_t, \widehat{\mathbf{y}}'_t)' := (\mathbf{g}'_t, y_{t-1}, \dots, y_{t-p})' := (1, \mathbf{w}'_t, \dots, \mathbf{w}'_{t-q+1}, \mathbf{x}'_t, y_{t-1}, \dots, y_{t-p})', \quad \text{and}$$

$$\boldsymbol{\alpha}_* := (\boldsymbol{\lambda}'_*, \boldsymbol{\phi}'_*)' := (\alpha_*, \delta'_*, \boldsymbol{\gamma}'_*, \boldsymbol{\phi}'_*)',$$

such that $\boldsymbol{\alpha}_*$ is an interior point of a convex and compact parameter space $\mathcal{A}$.

Remark 2. (i) The regressor $\mathbf{z}_t$ contains components with different stationarity orders:

$$\mathbf{z}_t = \left(\underbrace{1, \mathbf{w}'_t, \dots, \mathbf{w}'_{t-q+1}}_{\text{stationary, I(0)}}, \underbrace{\mathbf{x}'_t}_{\text{integrated, I(1)}}, \underbrace{y_{t-1}, \dots, y_{t-p}}_{\text{integrated, I(1)}} \right)'.$$

This mixed structure is a key distinction from the stationary data environment assumed by [Yao and Li \(2014\)](#), where the regularity condition assumes a finite fourth moment for the regressors. Meanwhile, this moment condition fails for $\mathbf{x}_t$ and y_{t-j} because $n^{-1} \sum_{t=1}^n \|\mathbf{x}_t\|^4 \asymp n^2$. The separate asymptotic theories need to be applied to the stationary and integrated components.

(ii) Each element of $\hat{\mathbf{y}}_t$ is formed as follows:

$$y_{t-i} = \mu_* + \mathbf{x}'_t \boldsymbol{\beta}_* + \sum_{j=0}^{q-1} \mathbf{w}'_{t-j} \boldsymbol{\xi}_{i,j*} + k_{t,i}, \quad i = 1, 2, \dots, p, \quad (8)$$

where $\boldsymbol{\xi}_{i,j*} := -\boldsymbol{\beta}_*$, if $i > j$; and $\boldsymbol{\xi}_{i,j*} := -\sum_{\ell=j-i}^{\infty} \pi_{\ell*}$, otherwise. Furthermore,

$$k_{t,i} := \begin{cases} -\sum_{j=q-i}^{\infty} \mathbf{w}'_{t-i-j} \boldsymbol{\xi}_{0,j*} + \sum_{j=0}^{\infty} \rho_{j*} u_{t-i-j}, & \text{if } i \leq q; \\ -\sum_{j=0}^{i-q-1} \mathbf{w}'_{t-q-j} \boldsymbol{\beta}_* + \sum_{j=0}^{\infty} \mathbf{w}'_{t-i-j} \boldsymbol{\pi}_{j*} + \sum_{j=0}^{\infty} \rho_{j*} u_{t-i-j}, & \text{if } i > q, \end{cases}$$

For each i , $k_{t,i}$ is stationary and ergodic. We note that (8) is the lagged version of (4), and use it frequently to examine the asymptotic behavior of our estimator. We rewrite it more compactly as $\hat{\mathbf{y}}_t = \boldsymbol{\Gamma}'_* \mathbf{g}_t + \mathbf{k}_t$ by assuming

$$\boldsymbol{\Gamma}_* := \begin{bmatrix} \mu_* \otimes \mathbf{1}_{1 \times p} \\ \boldsymbol{\Xi}_* \\ \boldsymbol{\beta}_* \otimes \mathbf{1}_{1 \times p} \end{bmatrix} := \begin{bmatrix} \mu_* & \mu_* & \cdots & \mu_* \\ \boldsymbol{\xi}_{1,0*} & \boldsymbol{\xi}_{2,0*} & \cdots & \boldsymbol{\xi}_{p,0*} \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{\xi}_{1,q-1*} & \boldsymbol{\xi}_{2,q-1*} & \cdots & \boldsymbol{\xi}_{p,q-1*} \\ \boldsymbol{\beta}_* & \boldsymbol{\beta}_* & \cdots & \boldsymbol{\beta}_* \end{bmatrix} \quad \text{and} \quad \mathbf{k}_t := \begin{bmatrix} k_{t,1} \\ k_{t,2} \\ \vdots \\ k_{t,p-1} \\ k_{t,p} \end{bmatrix}. \quad (9)$$

Here, $\mathbb{E}[\mathbf{k}_t]$ is not necessarily zero. □

We now estimate the unknown parameters in (7) by MR. Specifically,

$$\hat{\boldsymbol{\alpha}}_n := (\hat{\boldsymbol{\lambda}}'_n, \hat{\boldsymbol{\phi}}'_n)' := (\hat{\alpha}_n, \hat{\boldsymbol{\delta}}'_n, \hat{\boldsymbol{\gamma}}'_n, \hat{\boldsymbol{\phi}}'_n)' := \arg \max_{\boldsymbol{\alpha} \in \mathcal{A}} Q_{nh}(\boldsymbol{\alpha})$$

as the MARDL parameter estimator, where n is the sample size;

$$Q_{nh}(\boldsymbol{\alpha}) := \frac{1}{n} \sum_{t=1}^n \phi_h(y_t - \mathbf{z}'_t \boldsymbol{\alpha});$$

and $\phi_h(\cdot) := \frac{1}{h} \phi((\cdot)/h)$ is a kernel density function symmetric around 0, such as the standard normal density function. For our simulations and empirical analysis, we assume $\phi(\cdot)$ to be the standard normal density function.

Remark 3. (i) The maximization of $Q_{nh}(\cdot)$ requires a numerical optimization procedure because its maximum has no closed-form expression. For this, we utilize the modal expectation-maximization

(MEM) algorithm proposed by [Li et al. \(2007\)](#). The MEM algorithm comprises two steps, similarly to the EM algorithm. The E-step computes the weight of each observation. Given a parameter value, say $\alpha^{(g)}$, the posterior conditional probability of the t -th observation is computed as

$$\pi(t|\alpha^{(g)}) := \frac{\phi_h(y_t - \mathbf{z}'_t \alpha^{(g)})}{\sum_{i=1}^n \phi_h(y_i - \mathbf{z}'_i \alpha^{(g)})}.$$

Next, the M-step updates the parameter value using the weight calculated in the E-step by maximizing the following objective function:

$$\alpha^{(g+1)} := \arg \max_{\alpha \in \mathcal{A}} \sum_{t=1}^n \left\{ \pi(t|\alpha^{(g)}) \log[\phi_h(y_t - \mathbf{z}'_t \alpha)] \right\}.$$

[Yao and Li \(2014\)](#) demonstrate that $Q_{nh}(\alpha^{(g+1)}) > Q_{nh}(\alpha^{(g)})$, so if we continue these two steps until $\|\alpha^{(g)} - \alpha^{(g+1)}\|$ becomes less than a prespecified value, we can attain a value quite close to the maximum of $Q_{nh}(\cdot)$.

(ii) The bandwidth is set at $h = 1.6 \cdot n^{-1/7} \cdot \widehat{\text{MAD}}_n$, where

$$\widehat{\text{MAD}}_n := \text{med}_t \left\{ |\ddot{u}_t - \text{med}_s \ddot{u}_s| \right\}, \quad \text{where} \quad \ddot{u}_t := y_t - \mathbf{z}'_t \ddot{\alpha}_n,$$

with $\ddot{\alpha}_n$ a preliminary estimator for α_* in (7), e.g., a least-squares estimate. The first-stage scale estimate is required because the MARDL fit itself presupposes a bandwidth; the constant 1.6 is a pragmatic choice adopted by [Ullah et al. \(2023\)](#), following the MAD-based bandwidth in [Kemp and Silva \(2012\)](#). $\square$

We now examine the asymptotic behavior of the parameter estimator. For this, we provide the following regularized environment for the MARDL estimator along with the aforementioned assumptions:

Assumption 1. (i) $y_t = \mathbf{z}'_t \alpha_* + u_t$ such that

(i.a) the parameter space $\mathcal{A}$ is convex and compact such that α_* is an interior point of $\mathcal{A}$; and

(i.b) $\{u_t\}$ is an IID sequence with PDF $f_u(\cdot)$ such that for some $\lambda \in (0, 1]$, $\mathbb{E}|u_t|^{2+\lambda} < \infty$.

(ii) The kernel $\phi(\cdot) : \mathbb{R} \mapsto \mathbb{R}^+$ is a PDF symmetric around zero such that

(ii.a) $\phi(\cdot)$ is four times continuously differentiable; for $j = 1, 2, 3, 4$, $\phi^{(j)}(\cdot)$ is bounded, absolutely integrable, and square-integrable; and for the λ in (i.b) and $j = 0, 1, 2, 3, 4$, $|t|^{4+\lambda} |\phi^{(j)}(t)| \rightarrow 0$ as $|t| \rightarrow \infty$; and

(ii.b) for the λ in (i.b), $\int |\phi^{(1)}(t)|^{2+\lambda} dt < \infty$ and $\int t^4 \phi(t) dt < \infty$.

(iii) The PDF of u_t , viz., $f_u(\cdot)$, satisfies:

(iii.a) $f_u(\cdot) \in \mathcal{C}^{(3)}(\mathbb{R})$, and $f_u(\cdot)$ and its first three derivatives are uniformly bounded;

(iii.b) $f_u(u) < f_u(0)$ for any $u \neq 0$; and

(iii.c) $f_u^{(2)}(0) < 0$, where $f_u^{(\ell)}(\cdot)$ denotes the ℓ -th order derivative of $f_u(\cdot)$.

(iv) $\mathbf{w}_t := \Delta \mathbf{x}_t \in \mathbb{R}^k$ ($k \in \mathbb{N}$) is a linear multivariate stationary and ergodic process such that

(iv.a) for $j = 1, \dots, k$, $\mathbb{E}[w_{tj}] = 0$ and $\mathbb{E}[|w_{tj}|^{2+\lambda}] < \infty$, where λ is given in (i.b);

(iv.b) the sequences $\{u_t\}$ and $\{\mathbf{w}_t\}$ are independent;

(iv.c) the components of $\mathbf{x}_t$ are not cointegrated;

(iv.d) $\mathbb{E}\|(\bar{\mathbf{w}}'_t, \mathbf{k}'_t)'\|^6 < \infty$, where $\bar{\mathbf{w}}_t := [\mathbf{w}'_t, \dots, \mathbf{w}'_{t-q+1}]'$; and

(iv.e) $\mathbb{E}[\mathbf{j}_t \mathbf{j}'_t]$ is positive definite, where $\mathbf{j}_t := (1, \bar{\mathbf{w}}'_t, \mathbf{k}'_t)'$.

(v) The roots of $(1 - \sum_{j=1}^p \phi_{j*} L^j)$ lie outside the unit circle, $\sum_{j=0}^{\infty} \|\pi_{j*}\| < \infty$, and for each $i = 0, 1, \dots, p$, $\sum_{j=0}^{\infty} \|\xi_{i,j*}\| < \infty$.

(vi) Let $u_{th} := u_t/h$, $m_h := \mathbb{E}[\phi^{(1)}(u_{th})]$, $\bar{\phi}^{(1)}(u_{th}) := \phi^{(1)}(u_{th}) - m_h$, and $\bar{\mathbf{k}}_t := \mathbf{k}_t - \mathbb{E}[\mathbf{k}_t]$. Then $\mathbf{B}_n(\cdot) \rightsquigarrow \mathcal{B}(\cdot) := [\mathcal{B}_w(\cdot)', \mathcal{B}_k(\cdot)', \mathcal{B}_\phi(\cdot), \mathcal{B}_{\phi \cdot w}(\cdot)', \mathcal{B}_{\phi \cdot k}(\cdot)']'$, where

$$\mathbf{B}_n(\cdot) := \frac{1}{\sqrt{n}} \sum_{t=1}^{\lfloor n(\cdot) \rfloor} \left[\bar{\mathbf{w}}'_t, \bar{\mathbf{k}}'_t, \frac{\bar{\phi}^{(1)}(u_{th})}{\sqrt{h}}, \frac{\bar{\phi}^{(1)}(u_{th})}{\sqrt{h}} \bar{\mathbf{w}}'_t, \frac{\bar{\phi}^{(1)}(u_{th})}{\sqrt{h}} \bar{\mathbf{k}}'_t \right]',$$

and $\mathcal{B}(\cdot)$ is a multivariate Gaussian process with $\mathbb{E}[\mathcal{B}(\cdot)] = \mathbf{0}$ and, for each $r, \tilde{r} \in (0, 1)$, $\mathbb{E}[\mathcal{B}(r)\mathcal{B}(\tilde{r})'] = \boldsymbol{\Omega}(r, \tilde{r}) := \lim_{n \rightarrow \infty} \mathbb{E}[\mathbf{B}_n(r)\mathbf{B}_n(\tilde{r})']$ is positive definite.

(vii) $h \rightarrow 0$, $nh^5 \rightarrow \infty$, and $nh^7 = O(1)$ as $n \rightarrow \infty$. □

Remark 4. (i) The conditions in Assumption 1 extend the regularity conditions for MR to the context of integrated series (see Kemp and Silva, 2012; Yao and Li, 2014; Cho et al., 2015; Ullah et al., 2022, 2023, among others). A key distinction from the stationary data environment in Yao and Li (2014) is in the moment condition. Their finite fourth moment condition is violated in the MARDL context, as $n^{-1} \sum_{t=1}^n \|\mathbf{x}_t\|^4 = O_{\mathbb{P}}(n^2)$. We provide the current conditions to resolve this issue by separating the asymptotic treatment of the stationary series from that of the integrated series.

(ii) Assumption 1 (ii) permits kernels with unbounded support, provided that the stated tail and integrability conditions hold. Particularly, the standard normal density satisfies Assumption 1 (ii). Assumption 1 (ii.b) with $\lambda > 0$ is imposed to satisfy the Lindeberg-Lyapunov central limit theorem (CLT) for the modal score.

(iii) Assumptions 1 (iii.a) and (iii.b) are imposed to ensure a well-separated unique global mode at zero. Together with the differentiability condition, they imply $f_u^{(1)}(0) = 0$. By contrast, the strict curvature

condition in Assumption 1 (iii.c) is *not* implied by unimodality. For example, $f_u(u) \propto \exp(-u^4)$ has a unique well-separated mode at zero, but $f_u^{(2)}(0) = 0$. We impose Assumption 1 (iii.c) for model identification: $f_u^{(2)}(0) < 0$ is a key curvature condition and plays a role analogous to the quantile density $f_\tau := f(F^{-1}(\tau))$ in QARDL. Furthermore, the mode is well separated by Assumptions 1 (iii.a) and (iii.b): for any $\varepsilon > 0$, $\sup_{|u| \geq \varepsilon} f_u(u) < f_u(0)$. Indeed, $f_u^{(1)}(\cdot)$ is uniformly bounded by Assumption 1 (iii.a), so that $f_u(\cdot)$ is Lipschitz continuous and hence uniformly continuous. Furthermore, $f_u(\cdot)$ is integrable as it is a PDF, so that it vanishes at infinity because a uniformly continuous integrable function vanishes at infinity. Therefore, there is $M_\varepsilon < \infty$ such that $f_u(u) < f_u(0)/2$ whenever $|u| > M_\varepsilon$, whereas the continuous function $f_u(\cdot)$ attains its supremum on the compact set $\{\varepsilon \leq |u| \leq M_\varepsilon\}$, which is strictly less than $f_u(0)$ by Assumption 1 (iii.b).

- (iv) Assumption 1 (iv) imposes stationarity and ergodicity conditions on $\mathbf{w}_t := \Delta \mathbf{x}_t$. Under the IID condition in Assumption 1 (i.b) and the independence condition in Assumption 1 (iv.b), the conditional density of u_t on $\mathcal{F}_{t-1}$, viz., $f_u(\cdot|\mathcal{F}_{t-1})$, coincides with $f_u(\cdot)$. Furthermore, it follows that $\mathbb{E}[\phi^{(1)}(u_{th})|\mathcal{F}_{t-1}] = m_h$, so that $\{\bar{\phi}^{(1)}(u_{th}), \mathcal{F}_t\}$ is a martingale difference array (MDA). We can derive its asymptotic behavior by Taylor's expansion. By Assumption 1 (iii) and the symmetry of $\phi(\cdot)$, it follows that

$$m_h = -\frac{h^4}{2}\mu_2 f_u^{(3)}(0) + o(h^4) \rightarrow 0 \quad \text{and}$$

$$\text{var}\left(\frac{\bar{\phi}^{(1)}(u_{th})}{\sqrt{h}}\right) = \frac{1}{h} \int [\phi^{(1)}(u/h)]^2 f_u(u) du - \frac{m_h^2}{h} \rightarrow v^2 f_u(0),$$

as $h \rightarrow 0$, where $v^2 := \int [\phi^{(1)}(t)]^2 dt$ and $\mu_2 := \int t^2 \phi(t) dt$ such that $\phi^{(1)}(t) = -t\phi(t)$, $v^2 = 1/(4\sqrt{\pi})$, and $\mu_2 = 1$ for the standard normal kernel; see Section A.1 in the Online Supplement for this derivation. It follows that

$$\frac{1}{\sqrt{n}} \sum_{t=1}^n \frac{\bar{\phi}^{(1)}(u_{th})}{\sqrt{h}} = \frac{1}{\sqrt{nh}} \sum_{t=1}^n \left(\phi^{(1)}(u_{th}) - \mathbb{E}[\phi^{(1)}(u_{th})] \right) \overset{A}{\rightsquigarrow} N(0, v^2 f_u(0)),$$

We can extend this to the functional central limit theorem (FCLT), as follows:

$$\frac{1}{\sqrt{n}} \sum_{t=1}^{\lfloor n(\cdot) \rfloor} \frac{\bar{\phi}^{(1)}(u_{th})}{\sqrt{h}} \rightsquigarrow \mathcal{B}_\phi(\cdot).$$

- (v) Assumption 1 (v) is imposed for the existence of a stable long-run relationship between y_t and $\mathbf{x}_t$.
- (vi) By imposing the positive-definite matrix condition on $\mathbf{\Omega}(r, \tilde{r})$, we do not allow identical or constant sample paths among the processes in $\mathcal{B}(\cdot)$. Under Assumptions 1 (i.b) and (iv.b), every cross-covariance

between the $\bar{w}$ -components and the score components of $\mathbf{B}_n(\cdot)$ vanishes identically, so that $\mathbf{B}_w(\cdot)$, $\mathbf{B}_\phi(\cdot)$, and $\mathbf{B}_{\phi \cdot w}(\cdot)$ are mutually independent Gaussian processes. By contrast, $\mathbf{k}_s$ ($s \leq t$) is involved with lagged modal errors, so that the covariance between $\mathbf{B}_k(\cdot)$ and $\mathbf{B}_{\phi \cdot k}(\cdot)$ may not be zero.

- (vii) Let $\bar{\mathbf{B}}_w(\cdot)$ denote the first k elements of $\mathbf{B}_w(\cdot)$. By Assumptions 1 (iv) and (vi) and the continuous mapping theorem,

$$\frac{1}{n^2} \sum_{t=1}^n \mathbf{x}_t \mathbf{x}_t' \rightsquigarrow \int_0^1 \bar{\mathbf{B}}_w \bar{\mathbf{B}}_w', \quad \text{and} \quad \frac{1}{n\sqrt{h}} \sum_{t=1}^n \bar{\phi}^{(1)}(u_{th}) \mathbf{x}_t \rightsquigarrow \int_0^1 \bar{\mathbf{B}}_w d\mathbf{B}_\phi.$$

The latter follows from the MDA property of the centered score together with the fact that $\mathbf{x}_t$ is measurable with respect to $\mathcal{F}_{t-1}$; see Lemma A.4 (ii) in the Online Supplement. The independence condition in Assumption 1 (iv.b) further implies that the one-sided long-run covariance between the score and $\Delta \mathbf{x}_t$ is zero, so that its second-order bias term does not arise. The stochastic integral is mixed normal, as follows:

$$\int \bar{\mathbf{B}}_w d\mathbf{B}_\phi \sim N\left(\mathbf{0}, v^2 f_u(0) \int \bar{\mathbf{B}}_w \bar{\mathbf{B}}_w'\right)$$

conditional on $\bar{\mathbf{B}}_w(\cdot)$. This mixed normal distribution is one of the key ingredients to deriving the limit distribution of $\hat{\alpha}_n$.

- (viii) The independence condition in Assumption 1 (iv.b) is not as strong as it appears. The endogenous regressor problem can be removed by projecting the regression errors on $\Delta \mathbf{x}_t$ as detailed in [Cho et al. \(2015\)](#).
- (ix) The higher-level condition in Assumption 1 (vi) is attainable in a couple of ways. If $\{(\bar{w}_t', \bar{k}_t')'\}$ is an adapted mixingale of size -1 with uniformly bounded variance, we can apply the FCLT to it in a manner similar to [de Jong and Davidson \(2000\)](#) by noting the Lindeberg condition supplied in Assumption 1 (ii.b) together with Assumptions 1 (i.b) and (iv.a). Alternatively, one can strengthen the summability condition in Assumption 1 (v) by imposing a finite fourth moment condition. These two conditions do not necessarily imply each other, so we opt to impose the higher-level assumption.
- (x) The convergence rate of h in (vii) is deliberately provided so that the short- and long-run parameter estimators are asymptotically normal and do not suffer from smoothing bias. $\square$

For later use, we define the following limit processes before discussing the limit distribution of $\hat{\alpha}_n$: $\tilde{\mathbf{B}}_w(\cdot) :=$

$\bar{\mathcal{B}}_w(\cdot) - \int \bar{\mathcal{B}}_w$ and $\bar{\mathcal{B}}_\phi(\cdot) := \mathcal{B}_\phi(\cdot) - (\cdot) \mathcal{B}_\phi(1)$, which satisfy the following identity:

$$\int \bar{\mathcal{B}}_w d\bar{\mathcal{B}}_\phi = \int \tilde{\mathcal{B}}_w d\mathcal{B}_\phi \quad (10)$$

because $\int \tilde{\mathcal{B}}_w = \mathbf{0}$. The right side is mixed normal, viz., $\int \tilde{\mathcal{B}}_w d\mathcal{B}_\phi \sim N\left(\mathbf{0}, v^2 f_u(0) \int \tilde{\mathcal{B}}_w \tilde{\mathcal{B}}_w'\right)$ conditional on $\bar{\mathcal{B}}_w(\cdot)$.

2.1.3 Limit Behavior of the Estimator

We now derive the limit distribution of $\hat{\alpha}_n$. We apply a third-order Taylor expansion to the first-order condition for $\hat{\alpha}_n$ around α_* , and obtain

$$\mathbf{0} = -\frac{1}{nh^2} \sum_{t=1}^n \phi^{(1)}(u_{th}) \mathbf{z}_t + \frac{1}{nh^3} \sum_{t=1}^n \phi^{(2)}(u_{th}) \mathbf{z}_t \mathbf{z}_t' (\hat{\alpha}_n - \alpha_*) + \mathbf{r}_n, \quad (11)$$

where

$$\mathbf{r}_n := -\frac{1}{nh^4} \sum_{t=1}^n \left\{ \int_0^1 (1-s) \phi^{(3)}\left(u_{th} - s \frac{d_t}{h}\right) ds \right\} d_t^2 \mathbf{z}_t$$

such that $d_t := \mathbf{z}_t'(\hat{\alpha}_n - \alpha_*)$. Here, $\mathbf{r}_n$ is negligible relative to the first two terms on the right side; see Lemma A.5 in the Online Supplement.

Although the limit distribution of $\hat{\alpha}_n$ is obtainable by solving for it from (11), its derivation is not straightforward. The following two features of $\mathbf{z}_t$ prevent a direct analysis of (11). First, both stationary and integrated series exist in $\mathbf{z}_t$, so a common normalization does not apply. Second, the integrated series cannot be analyzed block by block, because every element of $\hat{\mathbf{y}}_t$ is cointegrated with $\mathbf{x}_t$ by (8), so multiple cointegrating relationships are estimated by $\hat{\alpha}_n$, making the blocks in the Hessian matrix formed by $\sum_{t=1}^n \phi^{(2)}(u_{th}) \mathbf{x}_t \mathbf{x}_t'$, $\sum_{t=1}^n \phi^{(2)}(u_{th}) \mathbf{x}_t \hat{\mathbf{y}}_t'$, and $\sum_{t=1}^n \phi^{(2)}(u_{th}) \hat{\mathbf{y}}_t \hat{\mathbf{y}}_t'$ be of order $n^2 h^3$. These two features make it difficult to obtain the limit distribution of $\hat{\alpha}_n$ from (11) directly.

We bypass both difficulties by reparameterization. Using $\hat{\mathbf{y}}_t = \mathbf{\Gamma}_*' \mathbf{g}_t + \mathbf{k}_t$ from (9), we note that

$$\mathbf{z}_t' \alpha_* = \mathbf{g}_t' \lambda_* + \hat{\mathbf{y}}_t' \phi_* = \mathbf{g}_t' (\lambda_* + \mathbf{\Gamma}_* \phi_*) + \mathbf{k}_t' \phi_* =: \mathbf{g}_t' \varsigma_\dagger + \mathbf{k}_t' \phi_\dagger \quad (12)$$

by letting $\varsigma_\dagger := \lambda_* + \mathbf{\Gamma}_* \phi_*$ and $\phi_\dagger = \phi_*$. If we decompose $\varsigma := \lambda + \mathbf{\Gamma}_* \phi$ into $(\nu, \mathbf{v}', \boldsymbol{\kappa}')'$ according to the structure of $\mathbf{g}_t$, then

$$\nu_\dagger = \alpha_* + \mu_*' \phi_*, \quad \mathbf{v}_\dagger = \boldsymbol{\delta}_* + \boldsymbol{\Xi}_* \phi_*, \quad \text{and} \quad \boldsymbol{\kappa}_\dagger = \boldsymbol{\gamma}_* + \boldsymbol{\beta}_* \sum_{j=1}^p \phi_{j*} = \boldsymbol{\beta}_*. \quad (13)$$

Therefore, the regressors in (12) are split into a stationary variable $\mathbf{j}_t := (1, \bar{\mathbf{w}}_t', \mathbf{k}_t')'$ with coefficient $\boldsymbol{\rho}_\dagger := (\nu_\dagger, \mathbf{v}_\dagger', \boldsymbol{\phi}_\dagger')'$, and the integrated regressor $\mathbf{x}_t$ with the long-run parameter $\boldsymbol{\beta}_*$. Using this separated structure, if $\boldsymbol{\vartheta} := (\boldsymbol{\rho}', \boldsymbol{\kappa}')' := (\nu, \mathbf{v}', \boldsymbol{\phi}', \boldsymbol{\kappa}')'$ and $\tilde{\mathbf{z}}_t := (\mathbf{j}_t', \mathbf{x}_t')'$, then $\mathbf{z}_t' \boldsymbol{\alpha}_* = \tilde{\mathbf{z}}_t' \boldsymbol{\vartheta}_\dagger$ by (12). Furthermore, from the definitions of $\boldsymbol{\varsigma}$ and $\boldsymbol{\rho}$, there exists a linear bijection relationship between $\boldsymbol{\alpha}$ and $\boldsymbol{\vartheta}$, so if $\tilde{Q}_{nh}(\cdot)$ is a function defined on the space of $\boldsymbol{\vartheta}$ such that for each $\boldsymbol{\alpha}$, $Q_{nh}(\boldsymbol{\alpha}) = \tilde{Q}_{nh}(\boldsymbol{\vartheta})$, then

$$\hat{\boldsymbol{\vartheta}}_n := (\hat{\boldsymbol{\rho}}_n', \hat{\boldsymbol{\kappa}}_n')' := (\hat{\nu}_n, \hat{\mathbf{v}}_n', \hat{\boldsymbol{\phi}}_n', \hat{\boldsymbol{\kappa}}_n')' := \arg \max_{\boldsymbol{\vartheta}} \tilde{Q}_{nh}(\boldsymbol{\vartheta})$$

can estimate the unknown parameter $\boldsymbol{\vartheta}_\dagger := (\boldsymbol{\rho}_\dagger', \boldsymbol{\kappa}_\dagger')'$ in parallel to $\hat{\boldsymbol{\alpha}}_n$. Due to the bijection relationship between $\boldsymbol{\alpha}$ and $\boldsymbol{\vartheta}$, the estimators also exhibit the following regular relationship:

$$\hat{\nu}_n = \hat{\alpha}_n + \mu_* \boldsymbol{\nu}_p' \hat{\boldsymbol{\phi}}_n, \quad \hat{\mathbf{v}}_n = \hat{\boldsymbol{\delta}}_n + \boldsymbol{\Xi}_* \hat{\boldsymbol{\phi}}_n, \quad \text{and} \quad \hat{\boldsymbol{\kappa}}_n = \hat{\boldsymbol{\gamma}}_n + \boldsymbol{\beta}_* \sum_{j=1}^p \hat{\boldsymbol{\phi}}_{jn}. \quad (14)$$

In addition, $\hat{\boldsymbol{\phi}}_n$ in $\hat{\boldsymbol{\alpha}}_n$ is identical to $\hat{\boldsymbol{\phi}}_n$ in $\hat{\boldsymbol{\vartheta}}_n$. By this reparameterization, the multiple cointegrating relationships between $\hat{\mathbf{y}}_t$ and $\mathbf{x}_t$ are converted into a single cointegrating relationship between y_t and $\mathbf{x}_t$. Exploiting this feature, we derive the limit distribution of $\hat{\boldsymbol{\alpha}}_n$ through the limit distribution of $\hat{\boldsymbol{\vartheta}}_n$.

Asymptotic Behavior of $\hat{\boldsymbol{\phi}}_n$. Although each y_{t-j} is $I(1)$, reparameterization (12) shows that its autoregressive parameter ϕ_* is now the coefficient of the stationary regressor $\mathbf{k}_t$. Furthermore, (13) implies that the $I(1)$ information carried by $\hat{\mathbf{y}}_t$ entirely disappears with the long-run relationship. By this simple fact, we can focus on the short-run variables to derive the limit distribution of $\hat{\boldsymbol{\phi}}_n$. Lemma A.6 (iii) in the Online Supplement obtains the following asymptotic approximation:

$$\begin{aligned} (\hat{\boldsymbol{\rho}}_n - \boldsymbol{\rho}_\dagger - b_h \mathbf{e}_1) &= \left(\sum_{t=1}^n \phi_t^{(2)}(u_{th}) \mathbf{j}_t \mathbf{j}_t' \right)^{-1} \sum_{t=1}^n \phi_t^{(1)}(u_{th}) \mathbf{j}_t \\ &\quad - \left(\sum_{t=1}^n \phi_t^{(2)}(u_{th}) \mathbf{j}_t \mathbf{j}_t' \right)^{-1} \left(\sum_{t=1}^n \phi_t^{(2)}(u_{th}) \mathbf{j}_t \mathbf{x}_t' \right) (\hat{\boldsymbol{\kappa}}_n - \boldsymbol{\beta}_*) + o_{\mathbb{P}}((nh^3)^{-1/2}), \end{aligned} \quad (15)$$

where $\mathbf{e}_1 := (1, \mathbf{0}_{(kq+p) \times 1}')'$ and $b_h := -\mu_2 h^2 f_u^{(3)}(0) / 2 f_u^{(2)}(0)$. Given this approximation, Lemmas A.6 (ii) and (iii) in the Online Supplement further show that

$$\sqrt{nh^3} \left(\sum_{t=1}^n \phi_t^{(2)}(u_{th}) \mathbf{j}_t \mathbf{j}_t' \right)^{-1} \sum_{t=1}^n \phi_t^{(1)}(u_{th}) \mathbf{j}_t \rightsquigarrow \boldsymbol{\ell}_\rho \sim B(\mathbf{0}, \boldsymbol{\Psi}^+),$$

$$\frac{1}{\sqrt{n}} \left(\sum_{t=1}^n \phi_t^{(2)}(u_{th}) \mathbf{j}_t \mathbf{j}_t' \right)^{-1} \left(\sum_{t=1}^n \phi_t^{(2)}(u_{th}) \mathbf{j}_t \mathbf{x}_t' \right) \rightsquigarrow \mathbf{e}_1 \left(\int \bar{\mathbf{B}}_w' \right), \quad \text{and}$$

$$n\sqrt{h^3}(\hat{\boldsymbol{\kappa}}_n - \boldsymbol{\beta}_*) \rightsquigarrow \boldsymbol{\ell}_{\boldsymbol{\kappa}} \sim N \left(\mathbf{0}, \omega_* \left[\int \tilde{\mathbf{B}}_w \tilde{\mathbf{B}}_w' \right]^{-1} \right)$$

conditional on $\tilde{\mathbf{B}}_w(\cdot)$, where $\boldsymbol{\Psi}^+ := \omega_* \mathbb{E}[\mathbf{j}_t \mathbf{j}_t']^{-1}$ and $\omega_* := v^2 f_u(0) / (f_u^{(2)}(0))^2$, so that it follows

$$\sqrt{nh^3}(\hat{\boldsymbol{\rho}}_n - \boldsymbol{\rho}_{\dagger} - b_h \mathbf{e}_1) \rightsquigarrow \boldsymbol{\ell}_{\boldsymbol{\rho}} - \mathbf{e}_1 \left(\int \bar{\mathbf{B}}_w' \right) \boldsymbol{\ell}_{\boldsymbol{\kappa}}.$$

Here, $\hat{\boldsymbol{\rho}}_n$ is obtained from minimizing $\tilde{Q}_n(\cdot)$, but $\hat{\boldsymbol{\phi}}_n$ in $\hat{\boldsymbol{\rho}}_n$ is exactly the same as $\hat{\boldsymbol{\phi}}_n$ minimizing $Q_n(\cdot)$. Therefore, the limit behavior of the latter is revealed by the limit distribution of the former. For this derivation, $\mathbf{s}_t := (\bar{\mathbf{w}}_t', \mathbf{k}_t')'$ so that $\mathbf{j}_t = (1, \mathbf{s}_t')'$, partitioning $\boldsymbol{\Psi}^+$ as follows:

$$\boldsymbol{\Psi}^+ := \begin{bmatrix} \psi_{00}^+ & \psi_{0s}^+ \\ \psi_{s0}^+ & \boldsymbol{\Psi} \end{bmatrix} \quad \text{such that} \quad \boldsymbol{\Psi} := \begin{bmatrix} \boldsymbol{\Psi}_{\bar{w}\bar{w}} & \boldsymbol{\Psi}_{\bar{w}\mathbf{k}} \\ \boldsymbol{\Psi}_{\mathbf{k}\bar{w}} & \boldsymbol{\Pi}_{p \times p} \end{bmatrix}. \quad (16)$$

Using the inverse matrix formula, it is straightforward to show that $\boldsymbol{\Psi} = \omega_* (\mathbb{E}[\mathbf{s}_t \mathbf{s}_t'] - \mathbb{E}[\mathbf{s}_t] \mathbb{E}[\mathbf{s}_t'])^{-1}$ and $\boldsymbol{\Pi} = \omega_* (\mathbb{E}[\mathbf{k}_t \mathbf{k}_t'] - \mathbb{E}[\mathbf{k}_t \hat{\mathbf{w}}_t'] \mathbb{E}[\hat{\mathbf{w}}_t \hat{\mathbf{w}}_t']^{-1} \mathbb{E}[\hat{\mathbf{w}}_t \mathbf{k}_t'])^{-1}$ with $\hat{\mathbf{w}}_t := (1, \bar{\mathbf{w}}_t')'$, so that $\boldsymbol{\Psi} = \omega_* [\text{var}(\mathbf{s}_t)]^{-1}$ and the asymptotic covariance matrix of $\hat{\boldsymbol{\phi}}_n$ is given as $\boldsymbol{\Pi}$. Furthermore, the asymptotic approximation in (15) can be concentrated with respect to $\hat{\boldsymbol{\phi}}_n$. If we let $\hat{\mathbf{h}}_t'$ be the t -th row vector of $\widehat{\mathbf{H}} := \mathbf{P}_{\widehat{\mathbf{W}}} \mathbf{K}$ with $\mathbf{P}_{\widehat{\mathbf{W}}} := \mathbf{I}_n - \widehat{\mathbf{W}} (\widehat{\mathbf{W}}' \widehat{\mathbf{W}})^{-1} \widehat{\mathbf{W}}'$ such that $\widehat{\mathbf{W}} := [\hat{\mathbf{w}}_1, \dots, \hat{\mathbf{w}}_n]'$ and $\mathbf{K} := [\mathbf{k}_1, \dots, \mathbf{k}_n]'$, then it follows that

$$\sqrt{nh^3}(\hat{\boldsymbol{\phi}}_n - \boldsymbol{\phi}_*) = \left(\frac{1}{nh^3} \sum_{t=1}^n \phi_t^{(2)}(u_{th}) \hat{\mathbf{h}}_t \hat{\mathbf{h}}_t' \right)^{-1} \frac{1}{\sqrt{nh}} \sum_{t=1}^n \bar{\phi}^{(1)}(u_{th}) \hat{\mathbf{h}}_t + o_{\mathbb{P}}(1) \stackrel{A}{\sim} N(\mathbf{0}, \boldsymbol{\Pi}), \quad (17)$$

from which the limit behavior of $\hat{\boldsymbol{\phi}}_n$ is obtained in parallel to Yao and Li (2014).

The following theorem contains the limit behavior:

Theorem 1. Given Assumption 1, $\sqrt{nh^3}(\hat{\boldsymbol{\phi}}_n - \boldsymbol{\phi}_*) \stackrel{A}{\sim} N(\mathbf{0}, \boldsymbol{\Pi})$. □

Remark 5. (i) If $\mathbf{h}_t := \mathbf{k}_t - \mathbb{E}[\mathbf{k}_t \hat{\mathbf{w}}_t'] \mathbb{E}[\hat{\mathbf{w}}_t \hat{\mathbf{w}}_t']^{-1} \hat{\mathbf{w}}_t$, it follows that $\boldsymbol{\Pi} = \omega_* \mathbb{E}[\mathbf{h}_t \mathbf{h}_t']^{-1}$ by noting that $\mathbf{h}_t = \hat{\mathbf{h}}_t + o_{\mathbb{P}}(1)$.

(ii) The independence condition between u_t and $\mathbf{j}_t$ given in Assumption 1 (i) plays a central role in deriving (17) from (15).

(iii) If $h \propto n^{-1/7}$, then $nh^5 = O(n^{2/7}) \rightarrow \infty$ and $nh^7 = O(1)$, so Assumption 1 (vii) holds. To motivate

the rate, the asymptotic mean squared error (AMSE) of the parameter estimator involves a variance with order $(nh^3)^{-1}$ and a squared bias with order h^4 , so minimizing the AMSE with respect to h yields an optimal bandwidth $h \propto n^{-1/7}$ (see Yao and Li, 2014; Ullah et al., 2022). Although this choice does *not* satisfy the undersmoothing condition $nh^7 \rightarrow 0$, (15) shows that the smoothing bias is associated only with the intercept estimation under Assumption 1 (iv.b), that is, all slope parameter estimators are free from the smoothing bias, and we can obtain their limit distribution using this optimal bandwidth.

- (iv) Smoothing bias does not arise in Theorem 1. Under Assumption 1 (iv.b), the conditional density of y_t on $\mathcal{F}_{t-1}$ is obtained from $f_u(\cdot)$ by shifting its location. Smoothing with bandwidth h perturbs the objective function identically at every t , and the common perturbation can shift only the location, namely, the intercept. Therefore, the pseudo-true values of all slope coefficients coincide with the true values. The smoothing bias associated with the intercept estimation is given in (15). As $b_h := -\frac{h^2}{2} \mu_2 f_u^{(3)}(0)/f_u^{(2)}(0)$, if $f_u^{(3)}(0) \neq 0$, then it follows that

$$\sqrt{nh^3}(\hat{\alpha}_n - \alpha_* - b_h) \rightsquigarrow s'_\alpha \ell_\rho - \left(\int \bar{\mathbf{B}}_w \right) \ell_\kappa, \quad \text{where} \quad s_\alpha := (1, \mathbf{0}'_{kq}, -\mu_* \boldsymbol{\iota}'_p)'$$

by noting that $\hat{\alpha}_n - \alpha_* = s'_\alpha(\hat{\boldsymbol{\rho}}_n - \boldsymbol{\rho}_*)$ and $s'_\alpha \mathbf{e}_1 = 1$; see Lemma A.7 (v) in the Online Supplement. Therefore, if $nh^7 = O(1)$, b_h asymptotically survives, but it vanishes if $nh^7 \rightarrow 0$. $\square$

Asymptotic Distribution of $\hat{\boldsymbol{\gamma}}_n$. For the asymptotic distribution of $\hat{\boldsymbol{\gamma}}_n, \hat{\boldsymbol{\varsigma}}_n := \hat{\boldsymbol{\lambda}}_n + \boldsymbol{\Gamma}_* \hat{\boldsymbol{\phi}}_n$ by following the definition of $\boldsymbol{\varsigma} := \boldsymbol{\lambda} + \boldsymbol{\Gamma}_* \boldsymbol{\phi}$. Furthermore, $\boldsymbol{\kappa}_* = \boldsymbol{\beta}_*$ by (13), so if we decompose $\hat{\boldsymbol{\varsigma}}_n$ into $(\hat{\nu}_n, \hat{\nu}'_n, \hat{\boldsymbol{\kappa}}'_n)'$ according to the structure of $\mathbf{g}_t$, then it follows that $\hat{\boldsymbol{\kappa}}_n - \boldsymbol{\beta}_* = (\hat{\boldsymbol{\gamma}}_n - \boldsymbol{\gamma}_*) + \boldsymbol{\beta}_* \boldsymbol{\iota}'_p (\hat{\boldsymbol{\phi}}_n - \boldsymbol{\phi}_*)$. Furthermore, Lemma A.6 (ii) in the Online Supplement obtains the asymptotic approximation of $\hat{\boldsymbol{\kappa}}_n - \boldsymbol{\beta}_*$ as follows:

$$\hat{\boldsymbol{\kappa}}_n - \boldsymbol{\beta}_* = \left[\frac{1}{n^2 h^3} \sum_{t=1}^n \phi^{(2)}(u_{th}) \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}'_t \right]^{-1} \left[\frac{1}{n^2 h^2} \sum_{t=1}^n \phi^{(1)}(u_{th}) \tilde{\mathbf{x}}_t \right] + o_{\mathbb{P}}((n\sqrt{h^3})^{-1}), \quad (18)$$

where $\tilde{\mathbf{x}}'_t$ is the t -th row vector of $\tilde{\mathbf{X}} := \mathbf{P}_{\hat{\mathbf{W}}} \mathbf{X}$. The limit distribution of $(\hat{\boldsymbol{\gamma}}_n - \boldsymbol{\gamma}_*)$ is obtained by combining (18) with (17). Given Theorem 1, we show that

$$\frac{1}{n^2 h^3} \sum_{t=1}^n \phi^{(2)}(u_{th}) \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}'_t \rightsquigarrow f_u^{(2)}(0) \int \tilde{\mathbf{B}}_w \tilde{\mathbf{B}}'_w \quad \text{and} \quad \frac{1}{n\sqrt{h}} \sum_{t=1}^n \phi^{(1)}(u_{th}) \tilde{\mathbf{x}}_t \rightsquigarrow \int \tilde{\mathbf{B}}_w d\mathbf{B}_\phi; \quad (19)$$

see Lemmas A.6 (i) and A.4 (v) in the Online Supplement. The second limit can also be written as $\int \bar{\mathbf{B}}_w d\bar{\mathbf{B}}_\phi$ by (10). Therefore, two convergence rates are in play: $\hat{\boldsymbol{\kappa}}_n - \boldsymbol{\beta}_* = O_{\mathbb{P}}((n\sqrt{h^3})^{-1})$ and $\hat{\boldsymbol{\phi}}_n - \boldsymbol{\phi}_* = O_{\mathbb{P}}((nh^3)^{-1/2})$, implying that the convergence rate of $\hat{\boldsymbol{\gamma}}_n$ is determined by that of $\hat{\boldsymbol{\phi}}_n$ because its convergence

rate is slower than that of $\hat{\kappa}_n$. Thus,

$$\sqrt{nh^3}(\hat{\gamma}_n - \gamma_*) = -\beta_* \iota'_p \sqrt{nh^3}(\hat{\phi}_n - \phi_*) + O_{\mathbb{P}}(n^{-1/2}). \quad (20)$$

By this and (18), the rate $n\sqrt{h^3}$ can be attached to $\hat{\kappa}_n - \beta_*$, so $n\sqrt{h^3}(\hat{\kappa}_n - \beta_*) \rightsquigarrow \ell_{\kappa}$ as stated above.

We provide the limit distributions of the estimators in the following theorem:

Theorem 2. *Given Assumption 1,*

- (i) $n\sqrt{h^3}\{(\hat{\gamma}_n - \gamma_*) + \beta_* \iota'_p(\hat{\phi}_n - \phi_*)\} \rightsquigarrow \ell_{\kappa} \sim N\left(\mathbf{0}, \omega_* [\int \tilde{\mathbf{B}}_w \tilde{\mathbf{B}}_w']^{-1}\right)$ conditional on $\bar{\mathbf{B}}_w(\cdot)$;
- (ii) $\sqrt{nh^3}(\hat{\gamma}_n - \gamma_*) \overset{A}{\sim} N\left(\mathbf{0}, \sigma_{\phi}^2 \beta_* \beta_*'\right)$, where $\sigma_{\phi}^2 := \iota'_p \Pi \iota_p$. □

Remark 6. (i) The sum of the rows in $P_{\widehat{W}}X$ is identically zero by construction. This implies that the score in (18) is invariant to the recentering of $\phi^{(1)}(u_{th})$, and the smoothing bias associated with the intercept estimation does not influence the score.

- (ii) Theorem 2 (ii) implies that Wald inference on γ_* is nonstandard because the rank of the asymptotic covariance matrix is one. We revisit this specifically in Section 2.2. □

Asymptotic Distribution of $\hat{\beta}_n$. To derive the asymptotic distribution of $\hat{\beta}_n$, we exploit the following:

$$\hat{\beta}_n - \beta_* = \left[1 - \sum_{j=1}^p \hat{\phi}_{j,n}\right]^{-1} \left[(\hat{\gamma}_n - \gamma_*) + \beta_* \iota'_p(\hat{\phi}_n - \phi_*)\right] = \left[1 - \sum_{j=1}^p \hat{\phi}_{j,n}\right]^{-1} (\hat{\kappa}_n - \beta_*). \quad (21)$$

which follows from (5), so if we combine (18) with (21),

$$n\sqrt{h^3}(\hat{\beta}_n - \beta_*) = \left[1 - \sum_{j=1}^p \hat{\phi}_{j,n}\right]^{-1} \left[\frac{1}{n^2 h^3} \sum_{t=1}^n \phi^{(2)}(u_{th}) \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t'\right]^{-1} \left[\frac{1}{n\sqrt{h}} \sum_{t=1}^n \phi^{(1)}(u_{th}) \tilde{\mathbf{x}}_t\right] + o_{\mathbb{P}}(1).$$

Here, $\sum_{j=1}^p \hat{\phi}_{j,n} \xrightarrow{\mathbb{P}} \sum_{j=1}^p \phi_{j*}$ by Theorem 1, and we can apply (19) to obtain the limit distribution of $\hat{\beta}_n$.

The following theorem provides the limit distribution:

Theorem 3. *Given Assumption 1,*

$$n\sqrt{h^3}(\hat{\beta}_n - \beta_*) \overset{A}{\sim} N\left(\mathbf{0}, \frac{\omega_*}{[1 - \sum_{j=1}^p \phi_{j*}]^2} \left[\int \tilde{\mathbf{B}}_w \tilde{\mathbf{B}}_w'\right]^{-1}\right)$$

conditional on $\bar{\mathbf{B}}_w(\cdot)$. □

Asymptotic Distribution of $\tilde{\varphi}_n$. The short-run autoregressive parameters in the EC representation are recovered from ϕ_* via (6): $\varphi_* = -\mathbf{R}_1\phi_*$. The short-run estimator in the EC representation is $\tilde{\varphi}_n := -\mathbf{R}_1\hat{\phi}_n$ based on the plug-in principle. The following theorem delivers its limit behavior:

Theorem 4. Given Assumption 1, $\sqrt{nh^3}(\tilde{\varphi}_n - \varphi_*) \overset{A}{\sim} N(\mathbf{0}, \mathbf{R}_1\mathbf{\Pi}\mathbf{R}_1')$. □

The covariance matrix $\mathbf{R}_1\mathbf{\Pi}\mathbf{R}_1'$ is positive definite because $\mathbf{R}_1$ has full row rank.

Asymptotic Distribution of $\tilde{\psi}_n$. If $\mathbf{S} := [\mathbf{I}_k, \mathbf{0}_{k \times k(q-1)}]'$, $\psi_* = \delta_* + \mathbf{S}\gamma_*$ by (6), we estimate ψ_* by $\tilde{\psi}_n := \hat{\delta}_n + \mathbf{S}\hat{\gamma}_n$. Furthermore, (12) gives $\mathbf{v}_\dagger = \delta_* + \mathbf{\Xi}_*\phi_*$, so that (20) and $\hat{\delta}_n - \delta_* = (\hat{\mathbf{v}}_n - \mathbf{v}_\dagger) - \mathbf{\Xi}_*(\hat{\phi}_n - \phi_*)$ delivers

$$\sqrt{nh^3}(\tilde{\psi}_n - \psi_*) = [\mathbf{I}_{kq} \ : \ -\mathbf{C}] \sqrt{nh^3} \begin{bmatrix} \hat{\mathbf{v}}_n - \mathbf{v}_\dagger \\ \hat{\phi}_n - \phi_* \end{bmatrix} + O_{\mathbb{P}}(n^{-1/2}), \quad \text{where } \mathbf{C} := \mathbf{\Xi}_* + \mathbf{S}\beta_*\iota'_p,$$

because $\hat{\kappa}_n - \beta_* = O_{\mathbb{P}}((n\sqrt{h^3})^{-1})$. Therefore, all components of $\tilde{\psi}_n$ converge at the rate of $\sqrt{nh^3}$. Therefore, (15) implies that

$$\sqrt{nh^3} \begin{bmatrix} \hat{\mathbf{v}}_n - \mathbf{v}_\dagger \\ \hat{\phi}_n - \phi_* \end{bmatrix} = \left(\frac{1}{nh^3} \sum_{t=1}^n \phi^{(2)}(u_{th}) \hat{\mathbf{s}}_t \hat{\mathbf{s}}_t' \right)^{-1} \frac{1}{\sqrt{nh}} \sum_{t=1}^n \bar{\phi}^{(1)}(u_{th}) \hat{\mathbf{s}}_t + o_{\mathbb{P}}(1) \overset{A}{\sim} N(\mathbf{0}, \mathbf{\Psi}),$$

where $\hat{\mathbf{s}}_t := \mathbf{s}_t - n^{-1} \sum_{t=1}^n \mathbf{s}_t$, so we can obtain the limit distributions of $\hat{\delta}_n$ and $\tilde{\psi}_n$ using this asymptotic distribution.

The following theorem reports the limit distributions of the estimators:

Theorem 5. Given Assumption 1,

- (i) $\sqrt{nh^3}(\hat{\delta}_n - \delta_*) \overset{A}{\sim} N(\mathbf{0}, \mathbf{\Omega}_\delta)$, where $\mathbf{\Omega}_\delta := [\mathbf{I}_{kq} \ : \ -\mathbf{\Xi}_*] \mathbf{\Psi} [\mathbf{I}_{kq} \ : \ -\mathbf{\Xi}_*]'$;
- (ii) $\sqrt{nh^3}(\tilde{\psi}_n - \psi_*) \overset{A}{\sim} N(\mathbf{0}, \mathbf{\Omega}_\psi)$, where $\mathbf{\Omega}_\psi := [\mathbf{I}_{kq} \ : \ -\mathbf{C}] \mathbf{\Psi} [\mathbf{I}_{kq} \ : \ -\mathbf{C}]'$. □

Remark 7. (i) Both $\mathbf{\Omega}_\delta$ and $\mathbf{\Omega}_\psi$ are positive definite by construction.

- (ii) The limit distribution of $\tilde{\psi}_n$ is not a mixed-normal distribution but only a normal distribution. □

2.2 Parameter Inference

We now develop inference procedures using the estimators given in Section 2.1.

Testing the Long-Run Parameter β_* . We propose the following hypotheses on the long-run parameter β_* :

$$\mathbb{H}_0^{(1)} : \mathbf{Q}\beta_* = \mathbf{q} \quad \text{versus} \quad \mathbb{H}_1^{(1)} : \mathbf{Q}\beta_* \neq \mathbf{q},$$

where $\mathbf{Q}$ is an $r \times k$ matrix with $\text{rank}(\mathbf{Q}) = r$ and $\mathbf{q}$ is an $r \times 1$ vector. To test the hypotheses, we apply the Wald test principle by considering the following test statistic:

$$\mathcal{W}_n(\beta) := h^3 \left(\mathbf{Q}\hat{\beta}_n - \mathbf{q} \right)' \left(\hat{\omega}_n \mathbf{Q} \left[1 - \sum_{j=1}^p \hat{\phi}_{j,n} \right]^{-2} \left(\sum_{t=1}^n \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t' \right)^{-1} \mathbf{Q}' \right)^{-1} \left(\mathbf{Q}\hat{\beta}_n - \mathbf{q} \right),$$

where

$$\hat{\omega}_n := \frac{v^2 \hat{f}_{un}(0)}{[\hat{f}_{un}^{(2)}(0)]^2} \quad \text{such that} \quad \hat{f}_{un}^{(\ell)}(0) := \frac{1}{n\tilde{h}^{\ell+1}} \sum_{t=1}^n K^{(\ell)}\left(\frac{\hat{u}_t}{\tilde{h}}\right), \quad \ell = 0, 1, 2.$$

Here, $\hat{u}_t := y_t - \mathbf{z}_t' \hat{\alpha}_n$; $K(\cdot)$ is a kernel function; and $\tilde{h}$ is a pilot bandwidth chosen independently of h such that $\tilde{h} \rightarrow 0$, $n\tilde{h} \rightarrow \infty$, and $n\tilde{h}^5 \rightarrow \infty$. From this and the fact that $\max_t |\hat{u}_t - u_t| = o_{\mathbb{P}}(1)$ as $\hat{\alpha}_n$ is consistent for α_* , it follows that $\hat{f}_u(0) \xrightarrow{\mathbb{P}} f_u(0)$ and $\hat{f}_u^{(2)}(0) \xrightarrow{\mathbb{P}} f_u^{(2)}(0)$ (see [Yao and Li, 2014](#), section 3.1). In addition, $v^2 := \int [\phi^{(1)}(t)]^2 dt$ is easily computed from the assumed kernel function, allowing $\hat{\omega}_n \xrightarrow{\mathbb{P}} \omega_*$. For the simulations and empirical illustrations in Sections 3 and 4, we let $K(\cdot)$ be the standard normal PDF and estimate $f_u(0)$ and $f_u^{(2)}(0)$ using the following pilot bandwidth:

$$\tilde{h} := 1.06n^{-\frac{1}{9}} \min[\hat{\sigma}_{\hat{u}}, \text{IQR}(\hat{u})/1.349],$$

where $\hat{\sigma}_{\hat{u}}$ and $\text{IQR}(\hat{u})$ denote the standard deviation and interquartile range of $\{\hat{u}_1, \dots, \hat{u}_n\}$, respectively, which lets $n\tilde{h}^5 = O_{\mathbb{P}}(n^{4/9})$. The rate $n^{-1/9}$ is the optimal bandwidth rate for estimating the second derivative of a density function (see [Härdle et al., 1990](#), for example), and the other scales are selected by the rule of thumb in [Silverman \(1986\)](#).

The Wald test is defined by exploiting the limit distribution of the long-run parameter estimator. Theorem 3 provides its limit distribution such that its limit covariance matrix is given by its sample analogue:

$$\frac{\hat{\omega}_n}{[1 - \sum_{j=1}^p \hat{\phi}_{j,n}]^2} \frac{1}{n^2} \sum_{t=1}^n \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t' \rightsquigarrow \frac{\omega_*}{[1 - \sum_{j=1}^p \phi_{j*}]^2} \int \tilde{\mathbf{B}}_w \tilde{\mathbf{B}}_w'.$$

Thus, it follows that $\mathcal{W}_n(\beta) \overset{A}{\sim} \chi_r^2$ under $\mathbb{H}_0^{(1)}$.

We present the limit behaviors of the Wald test in the following corollary:

Corollary 1. *Given Assumption 1, if $\hat{\omega}_n \xrightarrow{\mathbb{P}} \omega_*$ and $\text{rank}(\mathbf{Q}) = r$,*

- (i) $\mathcal{W}_n(\beta) \overset{A}{\sim} \chi_r^2$ under $\mathbb{H}_0^{(1)}$;
- (ii) $\mathbb{P}(\mathcal{W}_n(\beta) \geq s_n^{(1)}) \xrightarrow{\mathbb{P}} 1$ under $\mathbb{H}_1^{(1)}$, for any $s_n^{(1)} = o(n^2 h^3)$. □

Corollary 1 follows from Theorem 3 under the null and alternative. Unlike the level coefficient γ_* , the Wald test does not suffer from a singular matrix problem because the weight matrix is nonsingular with probability 1.

Testing the Short-Run Parameter ϕ_* . We propose the following hypotheses:

$$\mathbb{H}_0^{(2)} : \mathbf{R}\phi_* = \mathbf{r} \quad \text{versus} \quad \mathbb{H}_1^{(2)} : \mathbf{R}\phi_* \neq \mathbf{r},$$

where $\mathbf{R}$ is an $r \times p$ matrix with $\text{rank}(\mathbf{R}) = r$, and $\mathbf{r}$ is an $r \times 1$ vector. Based on Theorem 1, $\hat{\phi}_n$ is asymptotically normal; hence, the following is taken as the Wald test:

$$\mathcal{W}_n(\phi) := nh^3 \left(\mathbf{R}\hat{\phi}_n - \mathbf{r} \right)' \left(\mathbf{R}\hat{\Pi}_n \mathbf{R}' \right)^{-1} \left(\mathbf{R}\hat{\phi}_n - \mathbf{r} \right),$$

where $\hat{\Pi}_n$ is a consistent estimator for $\Pi := \omega_* \mathbb{E}[\mathbf{h}_t \mathbf{h}_t']^{-1}$. For example, a feasible estimator can be obtained by replacing Π with its sample analogue, say $\hat{\Pi}_n = \hat{\omega}_n (n^{-1} \sum_{t=1}^n \ddot{\mathbf{h}}_t \ddot{\mathbf{h}}_t')^{-1}$, where $\ddot{\mathbf{h}}_t := \hat{\mathbf{k}}_t - \sum_{t=1}^n \hat{\mathbf{k}}_t \hat{\mathbf{w}}_t' (\sum_{t=1}^n \hat{\mathbf{w}}_t \hat{\mathbf{w}}_t')^{-1} \hat{\mathbf{w}}_t$ such that $\hat{\mathbf{k}}_t := [\hat{k}_{t,1}, \dots, \hat{k}_{t,p}]'$, and for each $i = 1, \dots, p$, $\hat{k}_{t,i}$ is the residual obtained by regressing $y_{t-i} - \mathbf{x}_t' \hat{\beta}_n$ on $\hat{\mathbf{w}}_t = (1, \bar{\mathbf{w}}_t')'$, so that it consistently estimates the projection residual $\mathbf{h}_t$, letting $\hat{\Pi}_n \xrightarrow{\mathbb{P}} \Pi$ if $\hat{\omega}_n \xrightarrow{\mathbb{P}} \omega_*$ as discussed above.

We state the limit behaviors of $\mathcal{W}_n(\phi)$ as follows:

Corollary 2. *Given Assumption 1, if $\hat{\Pi}_n \xrightarrow{\mathbb{P}} \Pi$ and $\text{rank}(\mathbf{R}) = r$,*

- (i) $\mathcal{W}_n(\phi) \overset{A}{\sim} \chi_r^2$ under $\mathbb{H}_0^{(2)}$;
- (ii) $\mathbb{P}(\mathcal{W}_n(\phi) \geq s_n^{(2)}) \xrightarrow{\mathbb{P}} 1$ under $\mathbb{H}_1^{(2)}$, for any $s_n^{(2)} = o(nh^3)$. □

Corollary 2 follows from Theorem 1.

Testing the Short-Run Parameter γ_* . Inference on γ_* is fundamentally different from inference on β_* or ϕ_* . The main reason for this lies in the structural features of the modal cointegrating system. By Theorem 2 (ii), the rank of the asymptotic covariance matrix of $\hat{\gamma}_n$ is unity. Consequently, we cannot apply the standard Wald test principle to $\hat{\gamma}_n$ to test the following standard hypotheses:

$$\mathbb{H}_0^{(3)} : \mathbf{Q}\gamma_* = \mathbf{q} \quad \text{versus} \quad \mathbb{H}_1^{(3)} : \mathbf{Q}\gamma_* \neq \mathbf{q}.$$

Alternative testing procedures can be developed in a couple of ways. We note that

$$\mathbf{Q}(\hat{\gamma}_n - \gamma_*) = \mathbf{Q}(\hat{\kappa}_n - \beta_*) - (\mathbf{Q}\beta_*)' \iota_p'(\hat{\phi}_n - \phi_*) \quad (22)$$

by (18), and each component on the right side has a different convergence rate, as discussed earlier. Specifically, the convergence rate of $\hat{\phi}_n$ is slower than that of $\hat{\kappa}_n$. We can develop testing procedures exploiting both components on the right side separately. First, we suppose that $\mathbf{Q}\beta_* = \mathbf{0}$, so (22) reduces to $\mathbf{Q}(\hat{\gamma}_n - \gamma_*) = \mathbf{Q}(\hat{\kappa}_n - \beta_*)$, and the limit distribution of $(\hat{\kappa}_n - \beta_*)$ is given in Theorem 2 (i). Then, the Wald test is defined as follows:

$$\mathcal{W}_n^{(0)}(\gamma) := h^3 (\mathbf{Q}\hat{\gamma}_n - \mathbf{q})' \left(\hat{\omega}_n \mathbf{Q} \left[\sum_{t=1}^n \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t' \right]^{-1} \mathbf{Q}' \right)^{-1} (\mathbf{Q}\hat{\gamma}_n - \mathbf{q}).$$

Second, if $\mathbf{Q}\beta_* \neq \mathbf{0}$, the second term on the right side of (22) dominates, so it follows that $\sqrt{nh^3} \mathbf{Q}(\hat{\gamma}_n - \gamma_*) \stackrel{A}{\sim} N(\mathbf{0}, \sigma_\phi^2 (\mathbf{Q}\beta_*)(\mathbf{Q}\beta_*)')$ by Theorem 2 (ii). The rank of the asymptotic covariance matrix is unity, so defining a nondegenerate Wald test is attainable only when $\text{rank}(\mathbf{Q}) = 1$. We cannot test multiple restrictions jointly using the limit distribution of $\hat{\gamma}_n$. If the first row of $\mathbf{Q}$ and $\mathbf{q}$ is $\mathbf{q}_1'$ and q_1 , respectively, such that $\mathbf{q}_1' \beta_* \neq 0$, then a Wald test can be defined as follows:

$$\mathcal{W}_n^{(1)}(\gamma) := \frac{nh^3 (\mathbf{q}_1' \hat{\gamma}_n - q_1)^2}{\hat{\sigma}_{\phi n}^2 (\mathbf{q}_1' \hat{\beta}_n)^2}, \quad \text{where} \quad \hat{\sigma}_{\phi n}^2 := \iota_p' \hat{\Pi}_n \iota_p.$$

Here, $\hat{\Pi}_n$ is already defined when defining $\mathcal{W}_n(\phi)$. We collect the asymptotic behaviors of the Wald tests as follows:

Corollary 3. *Given Assumption 1,*

- (i) if $\mathbf{Q}\beta_* = \mathbf{0}$, $\text{rank}(\mathbf{Q}) = r$, and $\hat{\omega}_n \xrightarrow{\mathbb{P}} \omega_*$,
 - (i.a) $\mathcal{W}_n^{(0)}(\gamma) \stackrel{A}{\sim} \chi_r^2$ under $\mathbb{H}_0^{(3)}$;
 - (i.b) $\mathbb{P}(\mathcal{W}_n^{(0)}(\gamma) \geq s_n) \xrightarrow{\mathbb{P}} 1$ under $\mathbb{H}_1^{(3)}$ for any $s_n = o(n^2 h^3)$;
- (ii) if $\mathbf{Q}\beta_* \neq \mathbf{0}$, $\text{rank}(\mathbf{Q}) = 1$, and $\hat{\Pi}_n \xrightarrow{\mathbb{P}} \Pi$,
 - (ii.a) $\mathcal{W}_n^{(1)}(\gamma) \stackrel{A}{\sim} \chi_1^2$ under $\mathbb{H}_0^{(3)}$;
 - (ii.b) $\mathbb{P}(\mathcal{W}_n^{(1)}(\gamma) \geq s_n) \xrightarrow{\mathbb{P}} 1$ under $\mathbb{H}_1^{(3)}$ for any $s_n = o(nh^3)$. □

Remark 8. (i) $\mathbf{Q}\beta_* = \mathbf{0}$ or $\neq \mathbf{0}$ is a critical condition for applying Corollary 3. Before testing $\mathbb{H}_0^{(3)}$, we need to test $\mathbf{Q}\beta_* = \mathbf{0}$ using $\mathcal{W}_n(\beta)$.

(ii) The form of $\mathcal{W}_n^{(0)}(\gamma)$ is parallel to that of $\mathcal{W}_n(\beta)$. This follows from the identity $\gamma_* \equiv (1 -$

$\sum_{j=1}^p \phi_{j*})\beta_*$. Given that $\sum_{j=1}^p \phi_{j*} < 1$, imposing $\mathbf{Q}\gamma_* = \mathbf{0}$ is equivalent to $\mathbf{Q}\beta_* = \mathbf{0}$, so Corollary 3 implicitly assumes that $\mathbb{H}_0^{(3)} : \mathbf{Q}\beta_* = \mathbf{0}$, and $\mathcal{W}_n^{(0)}(\gamma)$ is defined by taking the weight matrix of $\mathcal{W}_n(\beta)$ as its weight matrix.

(iii) In the case $\mathbf{Q}\beta_* \neq \mathbf{0}$ but $\text{rank}(\mathbf{Q}) = r > 1$, then we may apply $\mathcal{W}_n^{(1)}(\gamma)$ by replacing $\mathbf{q}_1$ and q_1 with each row of $\mathbf{Q}$ and $\mathbf{q}$, respectively, so the r Wald tests can be applied separately. $\square$

Testing the Short-Run Parameter φ_* . To test the short-run autoregressive parameters $\varphi_* = -\mathbf{R}_1\phi_*$, we adopt the following hypotheses:

$$\mathbb{H}_0^{(4)} : \mathbf{N}\varphi_* = \mathbf{p} \quad \text{versus} \quad \mathbb{H}_1^{(4)} : \mathbf{N}\varphi_* \neq \mathbf{p},$$

where $\mathbf{N}$ is $r \times (p-1)$ with $\text{rank}(\mathbf{N}) = r \leq p-1$ and $\mathbf{p}$ is $r \times 1$. Since $\tilde{\varphi}_n = -\mathbf{R}_1\hat{\phi}_n$ is asymptotically normal by Theorem 4, we define the following Wald test:

$$\mathcal{W}_n(\varphi) := nh^3 (\mathbf{N}\tilde{\varphi}_n - \mathbf{p})' \left(\mathbf{N}\mathbf{R}_1\hat{\Pi}_n\mathbf{R}_1'\mathbf{N}' \right)^{-1} (\mathbf{N}\tilde{\varphi}_n - \mathbf{p}).$$

We present the asymptotic behaviors of the Wald test in the following corollary:

Corollary 4. *Given Assumption 1, if $\hat{\Pi}_n \xrightarrow{\mathbb{P}} \Pi$ and $\text{rank}(\mathbf{N}) = r$,*

- (i) $\mathcal{W}_n(\varphi) \overset{A}{\sim} \chi_r^2$ under $\mathbb{H}_0^{(4)}$;
- (ii) $\mathbb{P}(\mathcal{W}_n(\varphi) \geq s_n^{(4)}) \xrightarrow{\mathbb{P}} 1$ under $\mathbb{H}_1^{(4)}$, for any $s_n^{(4)} = o(nh^3)$. $\square$

As Corollary 4 follows from Theorem 4, we omit the proof.

Testing the Short-Run Parameter ψ_* . We test the following hypotheses on ψ_* :

$$\mathbb{H}_0^{(5)} : \mathbf{F}\psi_* = \mathbf{f} \quad \text{versus} \quad \mathbb{H}_1^{(5)} : \mathbf{F}\psi_* \neq \mathbf{f},$$

where $\mathbf{F}$ is $r \times (kq)$ with $\text{rank}(\mathbf{F}) = r$ and $\mathbf{f}$ is $r \times 1$. By Theorem 5 (iii), $\tilde{\psi}_n$ converges at the rate $\sqrt{nh^3}$ to a normal limit. Thus, the following Wald test tests these hypotheses:

$$\mathcal{W}_n(\psi) := nh^3 \left(\mathbf{F}\tilde{\psi}_n - \mathbf{f} \right)' \left(\mathbf{F}\hat{\Omega}_{\psi_n}\mathbf{F}' \right)^{-1} \left(\mathbf{F}\tilde{\psi}_n - \mathbf{f} \right),$$

where $\hat{\Omega}_{\psi_n}$ is a consistent estimator for $\Omega_\psi := [\mathbf{I}_{kq} \vdots -\mathbf{C}] \Psi [\mathbf{I}_{kq} \vdots -\mathbf{C}]'$. For example, a feasible estimator can be defined as $\hat{\Omega}_{\psi_n} := [\mathbf{I}_{kq} \vdots -\hat{\mathbf{C}}_n] \hat{\Psi}_n [\mathbf{I}_{kq} \vdots -\hat{\mathbf{C}}_n]'$. Here, $\hat{\mathbf{C}}_n := \hat{\Xi}_n + \mathbf{S}\hat{\beta}_n\iota_p'$ such that

$\widehat{\Psi}_n$ and $\widehat{\Xi}_n$ are constructed to estimate $\Psi := \omega_*(\mathbb{E}[\mathbf{s}_t \mathbf{s}_t'] - \mathbb{E}[\mathbf{s}_t] \mathbb{E}[\mathbf{s}_t']^{-1})^{-1}$ and Ξ_* , respectively, but neither of them is consistent for its target, because $\widehat{\mathbf{k}}_t$ is consistent for $\mathbf{h}_t$. Nevertheless, this inconsistency does not hinder the consistency of $\widehat{\Omega}_{\psi n}$. We allow

$$\widehat{\Psi}_n := \widehat{\omega}_n \left(\frac{1}{n} \sum_{t=1}^n \ddot{\mathbf{s}}_t \ddot{\mathbf{s}}_t' \right)^{-1} \quad \text{and} \quad \widehat{\Xi}_n := [\widehat{\xi}_{1,n}, \dots, \widehat{\xi}_{p,n}],$$

where $\ddot{\mathbf{s}}_t := (\overline{\mathbf{w}}_t', \widehat{\mathbf{k}}_t')'$, and for $i = 1, \dots, p$, $\widehat{\xi}_{i,n}$ is the OLS coefficient of $\overline{\mathbf{w}}_t$ that is obtained by regressing $y_{t-i} - \mathbf{x}_{t-i}' \widehat{\beta}_n$ on $\widehat{\mathbf{w}}_t := (1, \overline{\mathbf{w}}_t')'$.

We present the limit behaviors of the Wald test in the following corollary:

Corollary 5. *Given Assumption 1, if $\widehat{\Omega}_{\psi n} \xrightarrow{\mathbb{P}} \Omega_{\psi}$ and $\text{rank}(\mathbf{F}) = r$,*

- (i) $\mathcal{W}_n(\psi) \overset{A}{\sim} \chi_r^2$ under $\mathbb{H}_0^{(5)}$;
- (ii) $\mathbb{P}(\mathcal{W}_n(\psi) \geq s_n^{(5)}) \xrightarrow{\mathbb{P}} 1$ under $\mathbb{H}_1^{(5)}$, for any $s_n^{(5)} = o(nh^3)$. □

As Corollary 5 follows from Theorem 5, we omit the proof.

3 Monte Carlo Simulations

This section examines the properties of the Wald tests defined in Section 2.2 to corroborate the asymptotic theory on the estimators and test statistics.

3.1 Simulation Using MARDL Model

Focusing on estimating a MARDL model, we examine the following MARDL(1,1):

$$y_t = \alpha_{1*} + \phi_{1*} y_{t-1} + \theta_{0*} x_t + \theta_{1*} x_{t-1} + u_t,$$

where $x_t := x_{t-1} + w_t$ such that $w_t := \rho_* r_t + \sqrt{1 - \rho_*^2} r_{t-1}$ and $r_t \sim_{\text{iid}} U(-0.5, 0.5)$. We set the parameter values at $\alpha_{1*} = 0$, $\theta_{0*} = 0.5$, $\theta_{1*} = 0.5$, $\phi_{1*} = 0.5$, and $\rho_* = 0.5$ to examine the empirical rejection rates of the Wald tests under the null and alternative hypotheses. By construction, $\gamma_{1*} = 1$ and $\beta_{1*} = 2$. Given this, we estimate the unknown parameters in the MARDL(1,1) model, and further consider three distinct data-generating processes (DGPs) by assuming three different scenarios for u_t , which is assumed to be independent of x_t :

- DGP 1: $u_t \sim_{\text{iid}} N(0, 0.25)$, so that $\mathbb{E}(u_t) = 0$, $\text{median}(u_t) = 0$, and $\text{mode}(u_t) = 0$.
- DGP 2: $u_t \sim_{\text{iid}} \text{Beta}(2, 6) - 1/6$. By this, $\mathbb{E}(u_t) = 0.0833$, $\text{median}(u_t) = 0.0618$, and $\text{mode}(u_t) = 0$.

- DGP 3: $u_t \sim_{\text{iid}} \Gamma(3, 1) - 2$, so that $\mathbb{E}(u_t) = 1$, $\text{median}(u_t) = 0.6741$, and $\text{mode}(u_t) = 0$.

DGP 1 assumes a symmetric distribution for u_t , whereas DGPs 2 and 3 assume skewed distributions for u_t , so we can expect outliers more frequently from DGPs 2 and 3 than from DGP 1.

We examine the finite-sample performance of the Wald tests. Null simulation results are reported in Table 1. Hypothesis systems are specified as follows: first, the long-run parameter β_* is hypothesized as follows: $\mathbb{H}_0(\beta) : \beta_* = 2$ versus $\mathbb{H}_1(\beta) : \beta_* \neq 2$, which we test by $\mathcal{W}_n(\beta)$. Second, the short-run parameters ϕ_* and γ_* are hypothesized as follows: $\mathbb{H}_0(\phi) : \phi_* = 0.5$ versus $\mathbb{H}_1(\phi) : \phi_* \neq 0.5$, which we test by $\mathcal{W}_n(\phi)$. In addition, $\mathbb{H}_0(\gamma) : \gamma_* = 1$ versus $\mathbb{H}_1(\gamma) : \gamma_* \neq 1$. As $k = 1$ and $\beta_* \neq 0$ under $\mathbb{H}_0(\gamma)$, this null has to be tested by $\mathcal{W}_n^{(1)}(\gamma)$. Simulation results are obtained by repeating 3000 independent experiments, and we report the empirical rejection rates of $\mathcal{W}_n(\beta)$, $\mathcal{W}_n(\phi)$, and $\mathcal{W}_n^{(1)}(\gamma)$ in Table 1 for seven different sample sizes, $n = 200, 400, 600, 800, 1000, 2000$, and 3000 , when the significance level is 5%. As Table 1 shows, the empirical rejection rates of the tests converge to 5% as n grows. This corroborates Corollaries 1 (i), 2 (i), and 3 (ii.a).

Next, we report the power simulation results in Table 2. For the power simulation, we test $\mathbb{H}_0(\beta) : \beta_* = 1.95$, $\mathbb{H}_0(\phi) : \phi_* = 0.35$, and $\mathbb{H}_0(\gamma) : \gamma_* = 0.70$ against their corresponding two-sided alternatives. As before, we repeat the independent experiments 3000 times for $n = 200, 400, 600, 800, 1000, 2000, 3000$, and report the empirical rejection rates of $\mathcal{W}_n(\beta)$, $\mathcal{W}_n(\phi)$, and $\mathcal{W}_n^{(1)}(\gamma)$ when the significance level is 5%. Under DGPs 1 and 3, the rejection rates of $\mathcal{W}_n(\phi)$ and $\mathcal{W}_n^{(1)}(\gamma)$ grow from approximately 35–48% at $n = 200$ to approximately 97–99% at $n = 3000$. The power of $\mathcal{W}_n(\beta)$ also grows rapidly under DGP 1 but more slowly under DGP3, indicating that the finite-sample dispersion of u_t affects the power behavior. Under DGP 2, the powers grow more rapidly than under DGP 1 because the finite-sample dispersion of u_t is narrower. This corroborates Corollaries 1 (ii), 2 (ii), and 3 (ii.b).

We report additional simulation results in the Online Supplement. We examine the finite-sample biases and mean squared errors (MSEs) of the parameter estimators and compare their finite-sample distributions with their asymptotic distributions; see Section A.4.1 in the Online Supplement.

3.2 Simulation Using MARDL-EC Representation

We extend our simulation by considering the MARDL-EC representation. For this, we suppose the following MARDL(2, 1):

$$y_t = \alpha_{2*} + \phi_{2*}y_{t-1} + \phi_{3*}y_{t-2} + \theta_{2*}x_t + \theta_{3*}x_{t-1} + u_t,$$

where $x_t := x_{t-1} + w_t$, $w_t := \rho_* r_t + \sqrt{1 - \rho_*^2} r_{t-1}$, $r_t \sim_{\text{iid}} U(-0.5, 0.5)$. We set the parameter values at $\alpha_{2*} = 0$, $\theta_{2*} = 0.5$, $\theta_{3*} = 0.5$, $\phi_{2*} = 0.5$, $\phi_{3*} = 0$, and $\rho_* = 0.5$. Thus, MARDL-EC(2,1) is obtained as

$$\Delta y_t = \alpha_{3*} + \zeta_* (y_{t-1} - \beta_{2*} x_{t-1}) + \varphi_* \Delta y_{t-1} + \psi_* \Delta x_t + u_t$$

with $\gamma_{2*} = 1$ and $\beta_{2*} = 2$, and the short-run parameters are obtained as $\zeta_* = -0.5$, $\varphi_* = 0$, and $\psi_* = 0.5$. Given this, we assume the same three DGPs for u_t as in Section 3.1 and conduct simulations.

To examine the performance of the Wald tests defined by the MARDL-EC parameter estimators, we iterate 3000 independent experiments by considering $n = 200, 400, 600, 800, 1000, 2000$, and 3000 as before.

We report the null simulation results in Table 3, which reports the rejection rates at the significance level of 5%. We hypothesize the parameters as follows: $\mathbb{H}_0(\varphi) : \varphi_* = 0$ versus $\mathbb{H}_1(\varphi) : \varphi_* \neq 0$, which is tested by $\mathcal{W}_n(\varphi)$. We also let $\mathbb{H}_0(\psi) : \psi_* = 0.5$ versus $\mathbb{H}_1(\psi) : \psi_* \neq 0.5$ to be tested by $\mathcal{W}_n(\psi)$. All the empirical levels of the Wald tests trend toward the nominal level as n grows, implying that these tests are asymptotically well approximated by the chi-squared distribution with one degree of freedom. This corroborates Corollaries 4 (i) and 5 (i).

We report the power simulation results in Table 4. For the power examination, we let $\mathbb{H}_0(\varphi) : \varphi_* = -0.15$ versus $\mathbb{H}_1(\varphi) : \varphi_* \neq -0.15$, which is tested by $\mathcal{W}_n(\varphi)$. We further let $\mathbb{H}_0(\psi) : \psi_* = 0$ versus $\mathbb{H}_1(\psi) : \psi_* \neq 0$. The total number of experiments is 3000. The rejection rate of $\mathcal{W}_n(\varphi)$ gradually grows from 41.13%, 57.80%, and 35.33% at $n = 200$ to 91.50%, 99.63%, and 95.23% at $n = 3000$ under DGPs 1–3, respectively. The power of $\mathcal{W}_n(\psi)$ also increases with n , although the increase is relatively slow under DGP 3 because the finite-sample dispersion of u_t is greater than that under the other DGPs. This corroborates Corollaries 4 (ii) and 5 (ii).

Additional simulation results are reported in the Online Supplement. We examine the finite-sample biases and MSEs of the parameter estimators and compare their finite-sample distributions with their asymptotic distributions using QQ plots; see Section A.4.2 in the Online Supplement.

4 Empirical Application Using COVID-19 Data

We now apply the MARDL methodology to global COVID-19 data and extend its use to local data provided by four European countries.

4.1 Global Pandemic Data Analysis

We focus on global data analysis of daily COVID-19 confirmed cases and deaths.

4.1.1 Motivation and Model Set-Up

As [Fernández-Villaverde and Jones \(2020\)](#) point out, analyzing the COVID-19 pandemic and devising effective pandemic response measures are exceptionally challenging. When dealing with pandemic outbreaks, the primary task for scientists and policymakers is gaining a deep understanding of the pandemic trends and crisis evolution through both short- and long-run analyses.

Short- and long-run analyses are motivated by different objectives. Short-run predictive analysis plays a crucial role in resource allocation, immediate healthcare planning, and the formulation of interventions like social distancing and lockdown, as [Petroopoulos et al. \(2022\)](#) note. Meanwhile, [Ramazi et al. \(2021\)](#) point out that long-run trajectory analysis is frequently applied across various critical domains, including planning intensive care unit capacity, selecting clinical trial locations, ensuring ventilator supplies, and addressing vaccine production during pandemics, and even formulating national economic policy strategies.

While examining short- and long-run relationships between COVID-19 cases and deaths, we note the following two features. First, COVID-19 vaccination significantly and positively altered the course of the pandemic on a global scale. Besides the individual-level effects on disease risk and progression, vaccines also have secondary benefits in slowing down transmission rates and reducing related incidence and mortality rates. For example, prioritizing older individuals in vaccination proved the most effective in reducing the mortality rate (see [Dinleyici et al., 2021](#)).

Vaccination indeed offered protective and therapeutic effects against long-term COVID-19 and remains the cornerstone of the response to the pandemic, given the continued spread of the disease.

Despite the success of vaccination, achieving widespread public support for vaccination was challenging. Vaccine hesitancy and refusal continued partly due to misinformation, leading to suboptimal vaccination rates (see [Machingaidze and Wiysonge, 2021](#)). Some countries continued to rely on non-pharmaceutical interventions due to the significant heterogeneity in vaccination coverage. The reasons for vaccine hesitancy varied by context. [Lazarus et al. \(2023\)](#) identified more than 70 influencing factors, many of which are time- and context-specific, such as low education level, distrust in science and government, and misinformation.

Second, while global data analysis is key to the analysis of COVID-19, a global pandemic, it has been scarce. During the pandemic, the clinical and socio-economic impacts of safe and effective vaccines were significant to securing public health. However, empirical analyses of the clinical and economic effects of

rapidly deploying vaccines on a global scale are limited, although the impacts of these vaccines on public health and economic variables have been examined using local data (see [Di Fusco et al., 2022](#), for example). Only a few studies have employed empirical data to assess global public health impacts, particularly with limited data on the direct and indirect economic effects of COVID-19 vaccine administration (see [Yang et al., 2023](#), for example).

We contribute to the literature by focusing on both long- and short-run dynamic relationships between global COVID-19 confirmed cases and deaths. We apply MARDL methodology to investigate the short- and long-run dynamic relationships while considering the effects of vaccination. Compared to mean and median regressions for stationary series, MR is better for predicting periodic peaks and valleys more effectively because its variation range is broader and closer to the actual cyclic patterns. This feature is appealing even when examining integrated series. For example, [Ullah et al. \(2022\)](#) and [Jing and Cho \(2025\)](#) demonstrate that MR better fits COVID-19 data and produces forecasts more accurately than mean and median regressions.

Accordingly, we establish a model accommodating both long- and short-run relationships between the variables by applying the cointegration analysis of [Lintner's \(1956\)](#) dividend policy model to the current context (see also [Cho et al., 2015](#), section 5). First, we suppose that

$$\Delta d_t = \alpha_* - \zeta_*(d_t^* - d_{t-1}) + e_t,$$

where d_t and d_t^* represent the currently observed and predicted levels of deaths at time t , respectively. $|\zeta_*|$ is a smoothing coefficient between 0 and 1 that describes the degree of death smoothing. A long-run relationship between the number of COVID-19 deaths and confirmed cases is widely accepted, represented as $d_t^* = \beta_* c_t$, where β_* is the case fatality rate, and c_t denotes the level of confirmed cases at time t (see [Chen, 2022](#); [Petropoulos et al., 2022](#); [Jing and Cho, 2025](#), among others). We find that d_t and c_t are non-stationary, consistent with the empirical findings in [Ullah et al. \(2022\)](#) and [Jing and Cho \(2026\)](#). Therefore, $d_t^* = \beta_* c_t$ represents the cointegration relationship between d_t and c_t ,¹ obtaining

$$\Delta d_t = \alpha_* - \zeta_* \beta_* c_t + \zeta_* d_{t-1} + e_t = \alpha_* + \zeta_*(d_{t-1} - \beta_* c_{t-1}) + \omega_{0*} \Delta c_t + e_t,$$

where $\omega_{0*} := -\zeta_* \beta_*$. Importantly, there may be serial correlation in e_t . Therefore, we extend the model as follows:

$$\Delta d_t = \alpha_* + \zeta_* d_{t-1} + \gamma_* c_{t-1} + \sum_{j=1}^{p-1} \varphi_{j*} \Delta d_{t-j} + \sum_{j=0}^{q-1} \omega_{j*} \Delta c_{t-j} + \varepsilon_t.$$

¹These results are confirmed through unit root and cointegration tests, which are not reported here for brevity.

ε_t is assumed to have no serial correlation by letting the lag orders p and q be sufficiently large. However, note that Δc_t may be simultaneously correlated with the error term ε_t . For such a case, we can control the correlation by projecting ε_t onto Δc_t as follows: $\varepsilon_t = \nu_* \Delta c_t + u_t$, where u_t is constructed to be uncorrelated with Δc_t . Substituting this projection, we obtain the following ARDL(p, q):

$$\Delta d_t = \alpha_* + \zeta_*(d_{t-1} - \beta_* c_{t-1}) + \sum_{j=1}^{p-1} \varphi_{j*} \Delta d_{t-j} + \sum_{j=0}^{q-1} \psi_{j*} \Delta c_{t-j} + u_t, \quad (23)$$

where $\beta_* := -\frac{\gamma_*}{\zeta_*}$, $\psi_{0*} := \omega_{0*} + \nu_*$, and $\psi_{j*} := \omega_{j*}$ for $j = 1, \dots, q-1$.

Several remarks are in order for the parameters in (23). First, if $\nu_* \neq 0$, (23) cannot determine the simultaneous causality relationship between Δd_t and Δc_t . Otherwise, the changes in the number of COVID-19 cases can immediately cause the changes in the number of deaths under some mild regularity conditions, though the opposite is not true. Second, we can interpret the new coefficient ψ_{j*} on Δc_{t-j} as the impulse response parameter. Third, all other parameters, including β_* and ζ_* in (23), can be estimated by excluding endogeneity factors in the regression because Δc_t is included on the right side.

The following four parameters are of interest. First, the EC model parameter ζ_* measures the smoothing of mortality. Second, the long-run (cointegration) coefficient β_* measures the fatality rate. Third, $\varphi_* := \sum_{j=1}^{p-1} \varphi_{j*}$ measures the momentum effect of death, which quantifies the cumulative impact of past death growth on current growth, namely, $\sum_{j=1}^{p-1} \partial \Delta d_t / \partial \Delta d_{t-j}$. Finally, $\psi_* := \sum_{j=0}^{q-1} \psi_{j*}$ measures the impulse response of the increase in death to new confirmed cases, which quantifies the cumulative impact of the current and past growth of confirmed cases on the current death growth, viz., $\sum_{j=0}^{q-1} \partial \Delta d_t / \partial \Delta c_{t-j}$. Based on experience and theoretical analysis, our prior assumptions for the four parameters are as follows: $\zeta_* < 0$, $0 < \beta_* < 1$, $\varphi_* \leq 0$, and $\psi_* \geq 0$.

Using global pandemic data, we estimate the parameters of interest and investigate their implications in three dimensions. First, we estimate ARDL(p, q) using three different methods – ARDL, QARDL, and MARDL – and explore the implications targeted by each estimation. Second, we divide the timeline into pre- and post-vaccination periods to clearly compare the short- and long-run dynamic relationships between confirmed cases and deaths in order to explore the global effect of vaccination on public health. Finally, we investigate the evolving relationship between COVID-19 confirmed cases and deaths as vaccination becomes more widespread by the rolling window estimation method.

4.1.2 Empirical Analysis

We discuss the consequences of the analysis in terms of the three dimensions mentioned earlier. The global pandemic data and their source are described in Section A.5.1 of the Online Supplement. The sample ranges from December 8, 2020 to December 7, 2022, and Figure 1 presents the COVID-19 deaths and cases globally. We split the whole sample into two sub-periods: the pre-vaccination period from December 8, 2020 to December 7, 2021, and the post-vaccination period from December 8, 2021 to December 7, 2022. This sample division is based on the fact that the first dose of the vaccine is administered in a non-trial setting on December 8, 2020, and an estimated 55.9% of the global population has received at least one dose, 45.5% has received two, and 4.3% has received booster doses by December 8, 2021 (see Apergis, 2023). For model estimations, we select the orders p and q by the Bayesian Information Criterion (BIC): $p = 6$ and $q = 2$ for the pre-vaccination period, and $p = 7$ and $q = 1$ for the post-vaccination period.

A Comparison of the Three Model Estimations We specify the QARDL and MARDL models as counterparts of ARDL (p, q):

$$\Delta d_t = \alpha_*(\tau) + \zeta_*(\tau)d_{t-1} + \gamma_*(\tau)c_{t-1} + \sum_{j=1}^{p-1} \varphi_{j*}(\tau)\Delta d_{t-j} + \sum_{j=0}^{q-1} \psi_{j*}(\tau)\Delta c_{t-j} + u_t(\tau),$$

$$\Delta d_t = \alpha_*^M + \zeta_*^M d_{t-1} + \gamma_*^M c_{t-1} + \sum_{j=1}^{p-1} \varphi_{j*}^M \Delta d_{t-j} + \sum_{j=0}^{q-1} \psi_{j*}^M \Delta c_{t-j} + u_t^M.$$

Next, we estimate the ARDL, QARDL, and MARDL models by mean regression, quantile regression with the quantile level 0.5, and MR, respectively. Root MSE (RMSE), mean absolute error (MAE), and CQF results for the three ARDL methods are reported in Table 5. Here, RMSE is the root mean of the squared in-sample forecast errors, MAE is the mean of the absolute in-sample forecast errors, and CQF is the median of the absolute in-sample forecast errors.

We find that the MR method yields the smallest CQF value for the post-vaccination period, and the CQF values are more or less similar among the three estimation methods for the pre-vaccination period. The post-vaccination result indicates that the in-sample forecast obtained by MR captures the cyclical behavior of daily COVID-19 deaths more closely than those obtained by the mean and median regressions. When a time series exhibits multiple outliers and skewed distributions, the robustness and efficiency achieved by mean and median regression are compromised. Meanwhile, estimating the mode of the conditional distribution is not influenced by outliers or skewed distributions, allowing MR to track the cyclical behaviors of the time series

more successfully than mean and median regressions. Figure 2 illustrates the actual and fitted values of the global daily new deaths per million across the two vaccination periods.

Analysis of Vaccination Effects We examine the MARDL estimation results applied to the global data. The estimation results for the four parameters of interest are contained in the first panel of Table 6. We summarize the quantitative results as follows:

- (a) The EC coefficients ζ_*^M for the pre- and post-vaccination periods are -0.0292 and -0.1743, respectively, indicating adjustment rates of approximately 2.92% and 17.43%. This suggests that the system was returning to the long-run equilibrium state more rapidly, with the daily confirmed cases and daily deaths converging at a faster pace to culminate in their long-run relationship.
- (b) The long-run fatality rate β_*^M is 0.0571 during the pre-vaccination period, but decreases to 0.0036 during the post-vaccination period. The pre-vaccination ratio is weakly identified because its error-correction coefficient is close to zero, so that the magnitude of this decline needs to be carefully interpreted. Nevertheless, this direction indicates that the current deaths were less related to the confirmed cases following vaccination, which is consistent with the findings of [Watson et al. \(2022\)](#) who estimate a substantial reduction of approximately 63% in the potential global death toll attributable to vaccination.
- (c) The momentum effect parameters φ_*^M for growth of death cases are estimated to be -2.0323 and -4.2239 for the pre- and post-vaccination periods, respectively. This suggests that changes in newly confirmed death counts are negatively related to changes in past death counts, and the impact is negatively strengthened by vaccination, meaning that past deaths further reduce current deaths following vaccination.
- (d) The impact reaction parameter ψ_*^M of growth of death cases to the growth of confirmed cases reduced from 0.0096 to 0.0006 between the pre- and post-vaccination periods, implying that the association between the changes in deaths and cases has been weakened by vaccination. $\square$

Overall, the magnitudes and signs of the four key coefficients generally correspond with our initial assumptions, corroborating the vaccination effect.

Analysis by Rolling Window Approach We employ a robust rolling window estimation method and examine the evolution of the pandemic. We fix the window length at 266 days, and implement the rolling window method by iteratively sliding the estimation window forward by one day until reaching the end of the sample, producing 100 windows from each of the pre- and post-vaccination periods. We estimate

MARDL(6,2) and MARDL(7,1) using the samples in each window belonging to the pre- and post-vaccination periods, respectively. Figure 3 shows the evolution of the four parameter estimators obtained by the rolling window method. We summarize the estimation results as follows:

- (a) The rolling window estimation vividly illustrates significant temporal variation of the EC model coefficient ζ_* . Before vaccination, the adjustment speed fluctuates between -0.3% and -8.9%, but it ranges from -14% to -41% after vaccination and drifts towards the lower end as the window advances. These findings align with the overall analysis results.
- (b) The rolling window estimation for β_*^M reveals meaningful patterns. During the early stages of the pre-vaccination period, the long-run fatality rate shows an ascending trend reaching 0.38, followed by a substantial decrease and stabilization at around 0.03. This finding broadly aligns with the literature. In the initial months of 2021, responding to the need to establish infrastructure for vaccine distribution, some regions deferred the second dose, leading to a relatively minor impact of vaccination on the fatality rate. Furthermore, due to limited vaccine supply, many countries prioritized vaccination for high-risk individuals, including healthcare workers and the elderly. Such strategies yielded direct protective effects but had a relatively small impact on the spread of the pandemic. However, from mid-2021 onward, countries with ample vaccine supplies began large-scale vaccination campaigns for adults and children. Simultaneously, to maintain high-level protection, some regions also introduced booster shots. These measures contributed to lowering the fatality rate (see Imai et al., 2021; Wouters et al., 2021, for example). The rolling window mode estimates in the post-vaccination period exhibit a stabilizing pattern around 0.002.
- (c) Turning to the rolling mode estimates for φ_*^M , we observe fluctuating temporal patterns. During the early stages of the pre-vaccination period, φ_*^M displays a minor upward trend with negative values, ranging from approximately -2.3 to -1.7, implying that changes in the growth of past deaths were negatively associated with changes in current deaths. In the post-vaccination period, φ_*^M briefly moves between -2.5 and -4.2 before stabilizing around -3.6. This suggests that changes in the growth of past deaths were negatively related to changes in current deaths to a greater extent. This is because higher death counts in previous periods are linked to current death counts, thereby lowering current deaths. As vaccination became more effective, the current death count further decreased, and the negative correlation between the growth of past deaths and that of current deaths increased in magnitude.
- (d) Finally, during the pre-vaccination period, ψ_*^M gradually decreases from an early level of 0.015 to stabilize at around 0.010, after reaching a level as low as 0.009. In the post-vaccination period, ψ_*^M fluctuates in the early stages between 0.0005 and 0.0008, then stabilizes at around 0.0007. This

observation implies that, owing to vaccination, the correlation between changes in current deaths and changes in past and current confirmed cases weakened. □

4.2 Local Pandemic Data Analysis

We now focus on local data on COVID-19 confirmed cases and deaths in Belgium, Denmark, Germany, and Sweden. In addition to the four parameters, we analyze the effects of government response to the pandemic in these countries.

4.2.1 Motivation

Government response policies play a pivotal role in controlling disease transmission and reducing mortality rates. These measures include isolation, social distancing, mask usage, travel restrictions, and so on. Numerous studies have shown that lockdown and containment policies had a positive effect on reducing COVID-19 infections and deaths, particularly when implemented early (see [Pei et al., 2020](#); [Deb et al., 2022](#), among others).

Despite the widely recognized importance of interventions, the formulation and implementation of specific strategies against COVID-19 pose multiple challenges and uncertainties. For instance, a comprehensive consideration of factors is required when deciding when to initiate or lift restrictions and determining the intensity of interventions. These factors include public health, socio-economic considerations, public response, and the need to strike a balance between various interests and risks.

From this perspective, accumulating experience from previous pandemics is vital. We quantitatively compare our model estimates with government response policies on COVID-19 by utilizing the OxCGRT index to track governments' overall responses during the pre- and post-vaccination periods. This index reflects the strictness of policies related to measures such as lockdowns and economic policies. We particularly examine how the estimated relationships between confirmed cases and deaths vary across countries with lower and higher population densities and across different observed levels of government-response stringency during the pre- and post-vaccination periods.

For our empirical analysis, we select four European countries: Belgium, Denmark, Germany, and Sweden. We classify Belgium and Germany as one group, and Denmark and Sweden as another group by two complementary criteria: differences in policy stringency between the two groups and policy differences between the pre- and post-vaccination periods within each group. Specifically, government responses to COVID-19 in the four countries were not the same and also varied in the periods before and after vaccination. All European Union countries simultaneously started vaccination campaigns on December 27, 28, and 29,

2020. Figure 4 presents the overall government response index of the four countries. From the figures, we observe that each group exhibits different patterns, while the countries within the groups share similar patterns. Furthermore, during the pre- and post-vaccination periods, Sweden and Germany exhibited stricter measures compared to Denmark and Belgium, respectively. That is, averaged over each period, Germany has a stricter overall response index than Belgium: 63.54 against 51.72 before vaccination and 21.14 against 17.08 after vaccination. Likewise, Sweden is stricter than Denmark: 49.08 against 47.51 before vaccination and 15.74 against 14.69 after vaccination.

In addition, the countries within each group are broadly comparable in several observed economic, demographic, and health-system characteristics. Descriptive statistics related to COVID-19 are provided in Section A.5.2 in the Online Supplement. The two countries in each group share many similarities in terms of population density, per capita GDP, proportion of elderly population, smoking rate, and hospital bed capacity. From this, we can also classify Belgium and Germany as countries with higher population densities, and Denmark and Sweden as countries with relatively low population densities.

We investigate short- and long-run relationships between COVID-19 deaths and confirmed cases using MARDL. Figure 5 shows the time plots of COVID-19 deaths and cases in the four countries. As the four countries with similar counterparts implemented different response policies, we expect different estimation results among the four countries and between the periods before and after vaccination. In particular, the death series of the four countries are more dispersed, skewed, and outlier-prone than the global series, so that we can utilize this environment to compare conditional-mean, median, and mode substantively by applying MARDL along with ARDL and QARDL methodologies. From this examination, we shed light on the experiences of government response policies on COVID-19. All data used for our investigation are detailed in Section A.5.1 of the Online Supplement.

4.2.2 Empirical Analysis

Our local data analysis is conducted in parallel to the global data analysis. We first estimate ARDL, QARDL, and MARDL, and ensure the advantage of using MARDL over ARDL and QARDL. Next, we analyze the estimation results by focusing on the four parameters of interest.

Table 5 reports the RMSE, MAE, and CQF results obtained from the three models applied to the pre- and post-vaccination periods of the four countries. The analysis indicates that the MR method most often achieves the lowest CQF values across both periods, as in the global data analysis. This aspect is reinforced by comparing the fitted series of deaths with the actual series. Figures 6 and 7 present the in-sample forecasts of the two sets of countries, respectively, and the cyclical patterns of deaths are precisely tracked by MARDL.

We exploit MARDL to statistically analyze COVID-19 data and the implications of government responses.

We present the MARDL estimation results in Table 6 for the four countries across two distinct periods.

We summarize the quantitative results as follows:

- (a) The adjustment speeds ζ_*^M are negative for all four countries in both the pre- and post-vaccination periods and significant. In terms of their absolute values, the speed increases from 9.86% to 13.38% for Belgium and from 6.54% to 17.79% for Germany, as it does for the global data. Meanwhile, it is virtually unchanged for Denmark, from 13.33% to 12.89%, and for Sweden, the adjustment speed decreases from 10.39% to 8.06%. Therefore, the error correction speed after vaccination is faster than that before vaccination only for Belgium and Germany as for the global data, but not for Denmark and Sweden.
- (b) The success of overall government response policies should be assessed over the long run. We therefore compare the long-run fatality rate β_*^M of the four countries and compare the estimates with the government response indices during the pre- and post-vaccination periods.

First, we compare Belgium and Germany. During the pre-vaccination period the long-run fatality rates are 0.0044 for Belgium and 0.0087 for Germany, and during the post-vaccination period they fall to 0.0011 and 0.0015, respectively. Germany has the higher rate in both periods, although its mean response index is higher than Belgium for both periods. For instance, during the pre-vaccination period, Germany implemented lockdown measures starting from November 2020, which lasted until March 2021. Meanwhile, Belgium tightened its preventive measures, including border policies, starting from December 2020 and temporarily banned all non-essential travel from January 27 to April 1, 2021. In 2022, Germany maintained strict pandemic control policies and gradually eased the measures from February 2022 until all pandemic-related regulations were lifted in April 2023. Belgium began easing entry policies from March 11, 2022, transitioning from an orange to a yellow alert level. From May 23, 2022, the Belgian federal government officially suspended the COVID-19 alert level, further relaxing pandemic control measures and entry policies. Post-vaccination, Germany's fatality rate decreased to a level close to Belgium's. Their gap decreases from 0.0043 to 0.0004.

Second, we compare Denmark and Sweden. Sweden has the higher long-run fatality rate during the pre-vaccination period, 0.0051 against 0.0033 for Denmark, and the lower point estimate in the post-vaccination period, 0.0005 against 0.0009. The mean response indices of the two countries are close throughout, 49.08 for Sweden against 47.51 for Denmark before vaccination and 15.74 against 14.69 after. Denmark reinstated partial COVID-19 preventive measures at the end of November 2020 to control the COVID-19 spread. From March 2021, it gradually relaxed the measures in different

regions. Denmark re-introduced partial COVID-19 preventive measures at the end of November 2021 to address a new wave of infections during the winter. Denmark's government no longer classified COVID-19 as a critical societal threat after January 31, 2022, lifted all domestic pandemic restrictions, and revoked mandatory mask-wearing and vaccine passports on February 1, 2022, becoming the first EU country to remove all restrictions.

Taken together, the point estimates of the long-run fatality rate fall after vaccination in all four countries, which also aligns with the trends of the response indices. For Belgium and Germany, the country with a stricter mean response index has a higher estimated long-run fatality rate in both periods. The same ordering holds for Sweden and Denmark before vaccination, but it reverses afterward.

- (c) The momentum parameters φ_*^M of death growth are negative and significant at the 1% level for all four countries in both periods, so that a rise in the past change of deaths is followed by a fall in the current change. They become more negative after vaccination for Germany, from -2.1071 to -2.4025, and for Denmark, from -1.9354 to -2.3688, aligning with the global data. For Belgium and Sweden the estimates move in the opposite direction, from -1.5115 to -1.2481 and from -2.3665 to -0.7830, respectively. This shows that the momentum effect found for the global data is not uniform across local data.
- (d) The impact reaction of the change in deaths to the changes in confirmed cases ψ_*^M decreases after vaccination for Belgium, from 0.0005 to 0.0003, and for Germany, from 0.0148 to 0.0025, as it does for the global data. For the pre-vaccination period, the estimate of Denmark is indistinguishable from zero and its post-vaccination estimate is significantly negative at -0.0003. Meanwhile, that of Sweden falls from 0.0004 to 0.0001, and the latter is insignificant at the 5% level. For these two countries, the current change in deaths therefore shows no positive association with the contemporaneous change in confirmed cases, except for Denmark and Sweden after vaccination. $\square$

We close by summarizing the global and local analyses. First, MARDL broadly attains the smallest CQF for the global and local data in both the pre- and post-vaccination periods. This implies that the use of MR is appropriate for the analysis of COVID-19 data notably characterized by outliers. Second, the point estimates of the long-run fatality rate fall after vaccination for the global data and for all four countries, whereas the adjustment speed moves in the same direction as its global counterpart only for Belgium and Germany, and the momentum parameter only for Germany and Denmark. Furthermore, the within-group comparison shows that the estimated dynamics line up with the observed differences in response stringency.

5 Conclusion

The study makes a two-fold contribution. Theoretically, it provides a novel approach to cointegration analysis called the MARDL methodology that estimates a cointegrating relationship between integrated series by MR aimed at estimating the mode of the conditional distribution. The main motivation is that estimating short- and long-run relationships between integrated series using SLS is seriously affected by outliers. Therefore, we assume ARDL relationships between the variables and examine the asymptotic distribution of the parameter estimators. We further examine the asymptotic distributions of the estimators for the parameters in the EC representation implied by the ARDL model. In addition, we develop testing methodologies for the parameter estimators by applying the Wald test principle and corroborate the theoretical properties of the parameter estimators and test statistics by conducting extensive Monte Carlo simulations. In particular, the null limit distributions of the Wald tests defined in the present study are chi-squared, allowing for straightforward applicability.

An empirical application of the MARDL methodology is the second contribution. We examine cointegrating relationships between COVID-19 confirmed and death cases by applying the MARDL methodology to global data and local data from four European countries notably characterized by outliers, namely, Belgium, Denmark, Germany, and Sweden. Our interest lies in a quantitative assessment of the impact of vaccination and the role of government mitigation responses on the pandemic. Through the parameter estimates obtained by the MARDL methodology, we investigate how the pandemic evolved in response to vaccination and government pandemic mitigation efforts, and empirically affirm that the MARDL methodology broadly outperforms the forecasts delivered by ARDL and QARDL in terms of the CQF, which measures the median of the absolute in-sample forecast errors.

References

- APERGIS, N. (2023): “The connectedness between green and conventional bond yields during the COVID-19 crisis: the role of the vaccination process,” *Economics Letters*, 224, 111026. 28
- BERENGUER-RICO, V. AND B. NIELSEN (2026): “Least trimmed squares: cointegration and outliers,” *Oxford Bulletin of Economics and Statistics*, 88, 690–711. 1, 4
- BROCKWELL, P. J. AND R. A. DAVIS (1991): *Time Series: Theory and Methods*, Springer. 7
- CHEN, R.-M. (2022): “Analysing deaths and confirmed cases of COVID-19 pandemic by analytical approaches,” *European Physical Journal Special Topics*, 231, 3603–3617. 26

- CHEN, Y.-C. (2018): “Modal regression using kernel density estimation: a review,” *WIREs Computational Statistics*, 10, e1431. 2
- CHEN, Y.-C., C. R. GENOVESE, R. J. TIBSHIRANI, AND L. WASSERMAN (2016): “Nonparametric modal regression,” *Annals of Statistics*, 44, 489–514. 2
- CHO, J. S., T.-H. KIM, AND Y. SHIN (2015): “Quantile cointegration in the autoregressive distributed-lag modeling framework,” *Journal of Econometrics*, 188, 281–300. 1, 3, 9, 11, 26, 4, 5, 7
- COLLOMB, G., W. HÄRDLE, AND S. HASSANI (1986): “A note on prediction via estimation of the conditional mode function,” *Journal of Statistical Planning and Inference*, 15, 227–236. 2
- DE JONG, R. M. AND J. DAVIDSON (2000): “The Functional central limit theorem and weak convergence to stochastic integrals I: weakly dependent processes,” *Econometric Theory*, 16, 621–642. 11
- DEB, P., D. FURCERI, J. D. OSTRY, AND N. TAWK (2022): “The economic effects of COVID-19 containment measures,” *Open Economies Review*, 33, 1–32. 31
- DI FUSCO, M., K. MARCZELL, K. A. DEGER, M. M. MORAN, T. L. WIEMKEN, A. CANE, S. DE BOISVILLIERS, J. YANG, S. VAGHELA, AND J. ROIZ (2022): “Public health impact of the Pfizer-BioNTech COVID-19 vaccine (BNT162b2) in the first year of rollout in the United States,” *Journal of Medical Economics*, 25, 605–617. 26
- DINLEYICI, E. C., R. BORROW, M. A. P. SAFADI, P. VAN DAMME, AND F. M. MUNOZ (2021): “Vaccines and routine immunization strategies during the COVID-19 pandemic,” *Human Vaccines & Immunotherapeutics*, 17, 400–407. 25
- ENGLE, R. F. AND C. W. J. GRANGER (1987): “Co-integration and error correction: representation, estimation and testing,” *Econometrica*, 55, 251–276. 1, 4
- FAN, J. AND R. LI (2001): “Variable selection via nonconcave penalized likelihood and its oracle properties,” *Journal of the American Statistical Association*, 96, 1348–1360. 9
- FERNÁNDEZ-VILLAYERDE, J. AND C. I. JONES (2020): “Macroeconomic outcomes and COVID-19: a progress report,” Tech. rep., National Bureau of Economic Research. 25
- HAMILTON, J. D. (2020): *Time Series Analysis*, Princeton University Press. 7

- HANSEN, B. E. (1992): “Convergence to stochastic integrals for dependent heterogeneous processes,” *Econometric Theory*, 8, 489–500. 8
- HÄRDLE, W., J. S. MARRON, AND M. P. WAND (1990): “Bandwidth choice for density derivatives,” *Journal of the Royal Statistical Society: Series B (Methodological)*, 52, 223–232. 18
- IMAI, N., A. B. HOGAN, L. WILLIAMS, A. CORI, T. D. MANGAL, P. WINSKILL, L. K. WHITTLES, O. J. WATSON, E. S. KNOCK, M. BAGUELIN, ET AL. (2021): “Interpreting estimates of Coronavirus disease 2019 (COVID-19) vaccine efficacy and effectiveness to inform simulation studies of vaccine impact: a systematic review,” *Wellcome Open Research*, 6, 185. 30
- JING, X. AND J. S. CHO (2025): “Forecasting the confirmed COVID-19 cases using modal regression,” *Journal of Forecasting*, 44, 1578–1601. 2, 26
- (2026): “Quantile ARDL estimation of the relationship between confirmed COVID-19 cases and deaths in the U.S.” *Global Economic Review*, 55, 178–194. 26
- KEMP, G. C. R. AND J. M. C. S. SILVA (2012): “Regression towards the mode,” *Journal of Econometrics*, 170, 92–101. 2, 8, 9
- KRIEF, J. M. (2017): “Semi-linear mode regression,” *The Econometrics Journal*, 20, 149–167. 2
- LAZARUS, J. V., K. WYKA, T. M. WHITE, C. A. PICCHIO, L. O. GOSTIN, H. J. LARSON, K. RABIN, S. C. RATZAN, A. KAMARULZAMAN, AND A. EL-MOHANDES (2023): “A survey of COVID-19 vaccine acceptance across 23 countries in 2022,” *Nature Medicine*, 29, 366–375. 25
- LEE, M.-J. (1989): “Mode regression,” *Journal of Econometrics*, 42, 337–349. 2
- LEE, M.-J. (1993): “Quadratic mode regression,” *Journal of Econometrics*, 57, 1–19. 2
- LI, J., S. RAY, AND B. G. LINDSAY (2007): “A nonparametric statistical approach to clustering via mode identification,” *Journal of Machine Learning Research*, 8. 8
- LINTNER, J. (1956): “Distribution of incomes of corporations among dividends, retained earnings, and taxes,” *American Economic Review*, 46, 97–113. 26
- LYHAGEN, J. AND O. BARIS-TÜEMEN (2025): “Estimating cointegrating models in the presence of additive outliers,” *Communications in Statistics - Simulation and Computation*, forthcoming. 1, 4

- MACHINGAIDZE, S. AND C. S. WIYSONGE (2021): “Understanding COVID-19 vaccine hesitancy,” *Nature Medicine*, 27, 1338–1339. 25
- NIELSEN, H. B. (2004): “Cointegration analysis in the presence of outliers,” *The Econometrics Journal*, 7, 249–271. 1, 4
- OTA, H., K. KATO, AND S. HARA (2019): “Quantile regression approach to conditional mode estimation,” Tech. rep., arXiv:1712.08754. 2
- PEI, S., S. KANDULA, AND J. SHAMAN (2020): “Differential effects of intervention timing on COVID-19 spread in the United States,” *Science Advances*, 6, eabd6370. 31
- PESARAN, M. H. AND Y. SHIN (1998): “An autoregressive distributed lag modelling approach to cointegration analysis,” in *Econometrics and Economic Theory: The Ragnar Frisch Centennial Symposium*, ed. by S. Strom, Cambridge: Cambridge University Press, Econometric Society Monographs, 371–413. 1
- PETROPOULOS, F., S. MAKRIDAKIS, AND N. STYLIANOU (2022): “COVID-19: forecasting confirmed cases and deaths with a simple time series model,” *International Journal of Forecasting*, 38, 439–452. 25, 26
- PHILLIPS, P. AND S. DURLAUF (1986): “Multiple time series regression with integrated processes,” *Review of Economic Studies*, 53, 473–495. 7
- RAMAZI, P., A. HARATIAN, M. MEGHDADI, A. MARI ORIYAD, M. A. LEWIS, Z. MALEKI, R. VEGA, H. WANG, D. S. WISHART, AND R. GREINER (2021): “Accurate long-range forecasting of COVID-19 mortality in the USA,” *Scientific Reports*, 11, 13822. 25
- SAGER, T. W. AND R. A. THISTED (1982): “Maximum likelihood estimation of isotonic modal regression,” *Annals of Statistics*, 10, 690–707. 2
- SILVERMAN, B. W. (1986): *Density Estimation for Statistics and Data Analysis*, London: Chapman and Hall. 18
- ULLAH, A., T. WANG, AND W. YAO (2022): “Nonlinear modal regression for dependent data with application for predicting COVID-19,” *Journal of the Royal Statistical Society Series A: Statistics in Society*, 185, 1424–1453. 2, 9, 15, 26
- (2023): “Semiparametric partially linear varying coefficient modal regression,” *Journal of Econometrics*, 235, 1001–1026. 2, 8, 9

- WATSON, O. J., G. BARNSLEY, J. TOOR, A. B. HOGAN, P. WINSKILL, AND A. C. GHANI (2022): “Global impact of the first year of COVID-19 vaccination: a mathematical modelling study,” *Lancet Infectious Diseases*, 22, 1293–1302. 29
- WOUTERS, O. J., K. C. SHADLEN, M. SALCHER-KONRAD, A. J. POLLARD, H. J. LARSON, Y. TEERAWATTANANON, AND M. JIT (2021): “Challenges in ensuring global access to COVID-19 vaccines: production, affordability, allocation, and deployment,” *Lancet*, 397, 1023–1034. 30
- YANG, J., S. VAGHELA, B. YARNOFF, S. DE BOISVILLIERS, M. DI FUSCO, T. L. WIEMKEN, M. H. KYAW, J. M. MCLAUGHLIN, AND J. L. NGUYEN (2023): “Estimated global public health and economic impact of COVID-19 vaccines in the pre-Omicron era using real-world empirical data,” *Expert Review of Vaccines*, 22, 54–65. 26
- YANG, L., W. YUAN, AND S. WANG (2024): “Modal regression models based on B-splines,” *Computational Statistics*, 1–24. 2
- YAO, W. AND L. LI (2014): “A new regression model: modal linear regression,” *Scandinavian Journal of Statistics*, 41, 656–671. 2, 6, 8, 9, 14, 15, 18
- YAO, W., B. G. LINDSAY, AND R. LI (2012): “Local modal regression,” *Journal of Nonparametric Statistics*, 24, 647–663. 2
- YAO, W. AND S. XIANG (2016): “Nonparametric and varying coefficient modal regression,” Tech. rep., arXiv:1602.06609. 2
- ZHOU, H. AND X. HUANG (2019): “Bandwidth selection for nonparametric modal regression,” *Communications in Statistics-Simulation and Computation*, 48, 968–984. 2

DGP	Stat. \ n	200	400	600	800	1000	2000	3000
1	$\mathcal{W}_n(\phi)$	12.03	8.33	7.23	6.83	6.17	6.03	5.77
	$\mathcal{W}_n^{(1)}(\gamma)$	12.90	8.03	7.10	6.80	6.53	5.90	5.67
	$\mathcal{W}_n(\beta)$	10.73	8.03	6.57	6.67	7.57	5.77	5.67
2	$\mathcal{W}_n(\phi)$	10.80	7.57	5.93	7.63	7.23	5.47	6.73
	$\mathcal{W}_n^{(1)}(\gamma)$	10.30	7.57	5.90	7.20	6.93	5.60	6.57
	$\mathcal{W}_n(\beta)$	10.53	8.87	7.77	7.33	7.97	7.00	5.83
3	$\mathcal{W}_n(\phi)$	6.67	5.00	4.90	5.13	4.90	4.83	4.67
	$\mathcal{W}_n^{(1)}(\gamma)$	11.43	6.77	6.13	5.93	5.67	5.03	4.60
	$\mathcal{W}_n(\beta)$	7.83	5.80	5.07	4.73	4.90	4.40	4.37

Table 1: EMPIRICAL LEVELS OF $\mathcal{W}_n(\phi)$, $\mathcal{W}_n^{(1)}(\gamma)$, AND $\mathcal{W}_n(\beta)$ (LEVEL OF SIGNIFICANCE: 5%). Notes: (i) Number of replications: 3000. (ii) DGP: $y_t = \alpha_{1*} + \phi_{1*}y_{t-1} + \theta_{0*}x_t + \theta_{1*}x_{t-1} + u_t$, with $(\alpha_{1*}, \phi_{1*}, \theta_{0*}, \theta_{1*}) = (0, 0.5, 0.5, 0.5)$, $x_t = x_{t-1} + w_t$, $w_t = 0.5r_t + \sqrt{0.75}r_{t-1}$, and $r_t \sim_{\text{iid}} U(-0.5, 0.5)$. (iii) Null hypotheses: $\mathbb{H}_0^{(2)} : \phi_* = 0.5$, $\mathbb{H}_0^{(3)} : \gamma_* = 1$, and $\mathbb{H}_0^{(1)} : \beta_* = 2$. (iv) DGP 1: $u_t \sim_{\text{iid}} N(0, 0.25)$; DGP 2: $u_t \sim_{\text{iid}} \text{Beta}(2, 6) - 1/6$; DGP 3: $u_t \sim_{\text{iid}} \Gamma(3, 1) - 2$.

DGP	Stat. \ n	200	400	600	800	1000	2000	3000
1	$\mathcal{W}_n(\phi)$	44.10	56.53	65.03	71.93	78.60	92.83	97.20
	$\mathcal{W}_n^{(1)}(\gamma)$	48.43	62.13	70.70	76.93	81.27	93.37	97.20
	$\mathcal{W}_n(\beta)$	20.23	32.03	46.70	58.03	67.87	92.07	98.80
2	$\mathcal{W}_n(\phi)$	98.70	99.97	100.0	100.0	100.0	100.0	100.0
	$\mathcal{W}_n^{(1)}(\gamma)$	99.27	99.93	100.0	100.0	100.0	100.0	100.0
	$\mathcal{W}_n(\beta)$	70.97	93.63	98.40	99.93	100.0	100.0	100.0
3	$\mathcal{W}_n(\phi)$	34.60	52.23	63.70	72.67	79.17	94.43	98.50
	$\mathcal{W}_n^{(1)}(\gamma)$	40.53	55.20	66.10	73.47	80.00	93.93	97.50
	$\mathcal{W}_n(\beta)$	8.43	9.20	10.90	14.73	19.77	45.17	65.70

Table 2: EMPIRICAL POWERS OF $\mathcal{W}_n(\phi)$, $\mathcal{W}_n^{(1)}(\gamma)$, AND $\mathcal{W}_n(\beta)$ (LEVEL OF SIGNIFICANCE: 5%). Notes: (i) The number of replications, DGP, model, and error distributions are as in Table 1. (ii) Null hypotheses: $\mathbb{H}_0^{(2)} : \phi_* = 0.35$, $\mathbb{H}_0^{(3)} : \gamma_* = 0.70$, and $\mathbb{H}_0^{(1)} : \beta_* = 1.95$.

DGP	Stat. \ n	200	400	600	800	1000	2000	3000
1	$\mathcal{W}_n(\varphi)$	12.67	8.47	8.30	6.93	7.17	6.33	6.43
	$\mathcal{W}_n(\psi)$	13.73	9.33	8.83	7.70	7.43	6.27	5.37
2	$\mathcal{W}_n(\varphi)$	11.77	9.30	7.87	7.70	7.70	7.20	7.37
	$\mathcal{W}_n(\psi)$	11.73	8.40	8.93	7.73	8.07	7.53	7.13
3	$\mathcal{W}_n(\varphi)$	8.50	5.90	5.33	5.20	4.83	5.40	4.43
	$\mathcal{W}_n(\psi)$	8.90	6.93	5.30	5.20	4.17	4.60	4.77

Table 3: EMPIRICAL LEVELS OF $\mathcal{W}_n(\varphi)$ AND $\mathcal{W}_n(\psi)$ (LEVEL OF SIGNIFICANCE: 5%). Notes: (i) Number of replications: 3000. (ii) DGP: $y_t = \alpha_{2*} + \phi_{2*}y_{t-1} + \phi_{3*}y_{t-2} + \theta_{2*}x_t + \theta_{3*}x_{t-1} + u_t$, with $(\alpha_{2*}, \theta_{2*}, \theta_{3*}, \phi_{2*}, \phi_{3*}) = (0, 0.5, 0.5, 0.5, 0)$ and the process for x_t as in Table 1. (iii) The EC representation is $\Delta y_t = \alpha_{3*} + \zeta_*(y_{t-1} - \beta_{2*}x_{t-1}) + \varphi_*\Delta y_{t-1} + \psi_*\Delta x_t + u_t$. (iv) Null hypotheses: $\mathbb{H}_0^{(4)} : \varphi_* = 0$ and $\mathbb{H}_0^{(5)} : \psi_* = 0.5$. (v) The three error distributions are the same as given in Table 1.

DGP	Stat. \ n	200	400	600	800	1000	2000	3000
1	$\mathcal{W}_n(\varphi)$	41.13	50.20	57.17	62.27	66.00	83.87	91.50
	$\mathcal{W}_n(\psi)$	57.73	71.97	80.73	87.50	90.60	97.90	99.60
2	$\mathcal{W}_n(\varphi)$	57.80	75.50	83.90	89.10	92.90	98.80	99.63
	$\mathcal{W}_n(\psi)$	100.0	100.0	100.0	100.0	100.0	100.0	100.0
3	$\mathcal{W}_n(\varphi)$	35.33	47.63	57.67	66.30	72.40	90.23	95.23
	$\mathcal{W}_n(\psi)$	15.93	17.47	19.47	22.80	25.37	36.10	47.07

Table 4: EMPIRICAL POWERS OF $\mathcal{W}_n(\varphi)$ AND $\mathcal{W}_n(\psi)$ (LEVEL OF SIGNIFICANCE: 5%). Notes: (i) The number of replications, DGP, EC representation, and error distributions are as in Table 3. (ii) Null hypotheses: $\mathbb{H}_0^{(4)} : \varphi_* = -0.15$ and $\mathbb{H}_0^{(5)} : \psi_* = 0$.

Country	Period	Model	RMSE	MAE	CQF	(p, q)
World	Pre-vaccination	ARDL	0.1182	0.0650	0.0439	(6,2)
		QARDL	0.1196	0.0634	0.0429	
		MARDL	0.1216	0.0646	0.0444	
	Post-vaccination	ARDL	0.1025	0.0527	0.0307	(7,1)
		QARDL	0.1073	0.0487	0.0229	
		MARDL	0.1154	0.0502	0.0215	
Belgium	Pre-vaccination	ARDL	0.5141	0.3606	0.2432	(6,1)
		QARDL	0.5189	0.3567	0.2300	
		MARDL	0.5392	0.3706	0.2272	
	Post-vaccination	ARDL	0.4139	0.2962	0.2039	(4,1)
		QARDL	0.4164	0.2942	0.2064	
		MARDL	0.4194	0.2956	0.2069	
Germany	Pre-vaccination	ARDL	0.5191	0.3171	0.1878	(8,6)
		QARDL	0.5801	0.2797	0.1100	
		MARDL	0.6212	0.2899	0.0953	
	Post-vaccination	ARDL	0.2725	0.1913	0.1322	(9,8)
		QARDL	0.2905	0.1823	0.1176	
		MARDL	0.3069	0.1876	0.1044	
Denmark	Pre-vaccination	ARDL	0.5471	0.3151	0.1904	(5,1)
		QARDL	0.5511	0.3115	0.1813	
		MARDL	0.5896	0.3268	0.1647	
	Post-vaccination	ARDL	1.1454	0.6449	0.3726	(6,1)
		QARDL	1.1704	0.6222	0.3696	
		MARDL	1.1621	0.6316	0.3345	
Sweden	Pre-vaccination	ARDL	0.4490	0.2951	0.1774	(12,1)
		QARDL	0.4592	0.2855	0.1650	
		MARDL	0.5024	0.2993	0.1338	
	Post-vaccination	ARDL	0.4351	0.3320	0.2851	(3,1)
		QARDL	0.4418	0.3292	0.2681	
		MARDL	0.4507	0.3337	0.2641	

Table 5: THE QUANTITATIVE RESULTS OF DIFFERENT ESTIMATION METHODS. This table evaluates the efficacy of the three different ARDL methods using the COVID-19 data. The lags are determined by Bayesian Information Criterion (BIC). All three measures are computed using the in-sample forecasts. RMSE is the root mean of the squared in-sample forecast errors, MAE is the mean of the absolute in-sample forecast errors, and CQF is the median of the absolute in-sample forecast errors.

Country	Period	ζ_*^M	β_*^M	φ_*^M	ψ_*^M
World	Pre-vaccination	-0.0292*** (0.0076)	0.0571*** (0.0121)	-2.0323*** (0.1431)	0.0096*** (0.0013)
	Post-vaccination	-0.1743*** (0.0071)	0.0036*** (0.0001)	-4.2239*** (0.0586)	0.0006*** (0.0000)
Belgium	Pre-vaccination	-0.0986*** (0.0129)	0.0044*** (0.0008)	-1.5115*** (0.2199)	0.0005*** (0.0001)
	Post-vaccination	-0.1338*** (0.0274)	0.0011*** (0.0002)	-1.2481*** (0.1530)	0.0003*** (0.0001)
Germany	Pre-vaccination	-0.0654*** (0.0015)	0.0087*** (0.0004)	-2.1071*** (0.0308)	0.0148*** (0.0005)
	Post-vaccination	-0.1779*** (0.0128)	0.0015*** (0.0001)	-2.4025*** (0.1679)	0.0025*** (0.0003)
Denmark	Pre-vaccination	-0.1333*** (0.0133)	0.0033*** (0.0008)	-1.9354*** (0.1047)	0.0000 (0.0003)
	Post-vaccination	-0.1289*** (0.0173)	0.0009*** (0.0001)	-2.3688*** (0.1255)	-0.0003*** (0.0001)
Sweden	Pre-vaccination	-0.1039*** (0.0045)	0.0051*** (0.0004)	-2.3665*** (0.1375)	0.0004*** (0.0001)
	Post-vaccination	-0.0806* (0.0374)	0.0005 (0.0006)	-0.7830*** (0.1504)	0.0001 (0.0002)

Table 6: MARDL ESTIMATION RESULTS. Notes: (i) * indicates $p < 0.05$, ** indicates $p < 0.01$, and *** indicates $p < 0.001$. (ii) Figures in parentheses are standard errors obtained from the estimators of the asymptotic variance in Theorems 1–5. The significance symbols are based exclusively on the analytic standard errors reported in parentheses. (iii) MARDL estimation results are based on the following model: $\Delta d_t = \alpha_*^M + \zeta_*^M d_{t-1} + \gamma_*^M c_{t-1} + \sum_{j=1}^{p-1} \varphi_{j*}^M \Delta d_{t-j} + \sum_{j=0}^{q-1} \psi_{j*}^M \Delta c_{t-j} + u_t^M = \alpha_*^M + \zeta_*^M (d_{t-1} - \beta_*^M c_{t-1}) + \sum_{j=1}^{p-1} \varphi_{j*}^M \Delta d_{t-j} + \sum_{j=0}^{q-1} \psi_{j*}^M \Delta c_{t-j} + u_t^M$.

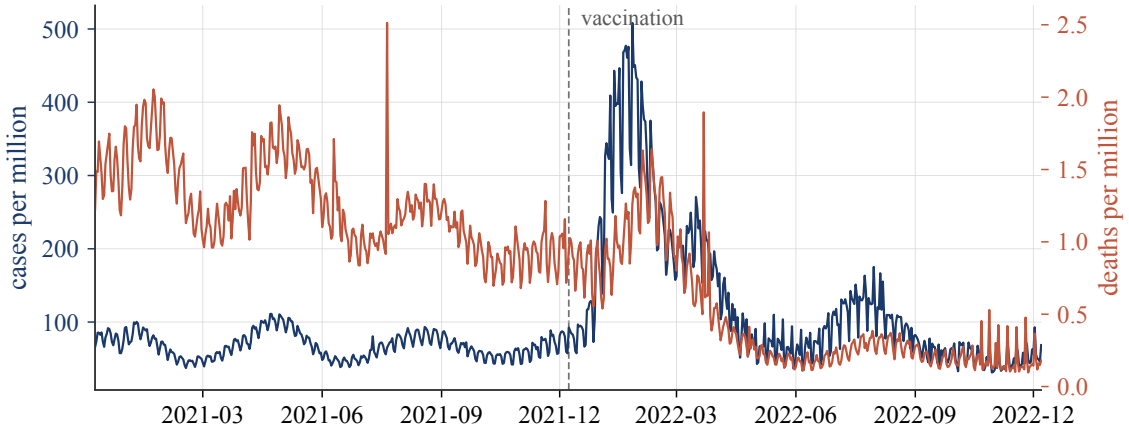


Figure 1: DAILY GLOBAL COVID-19 CASES AND DEATHS PER MILLION. Notes: (i) The sample ranges from December 8, 2020 to December 7, 2022. (ii) The left axis measures new cases per million in blue, and the right axis measures new deaths per million in red. (iii) The dashed vertical line at December 8, 2021 separates the pre- and post-vaccination periods. (iv) Source: Our World in Data; see Section A.5.1 in the Online Supplement.

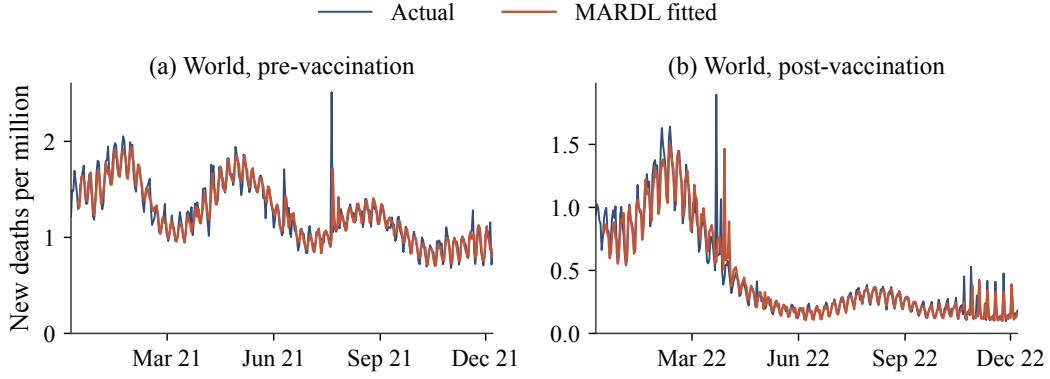


Figure 2: ACTUAL AND MARDL FITTED DAILY NEW DEATHS PER MILLION USING THE GLOBAL DATA. Notes: (i) The in-sample fitted values are drawn together with the actual values. (ii) Panel (a) covers the pre-vaccination period, and panel (b) covers the post-vaccination period. (iii) Lag orders are those in Table 5. (iv) The vertical axis measures new deaths per million people; fitted values are truncated below at zero. (v) Source: Our World in Data; see Section A.5.1 in the Online Supplement.

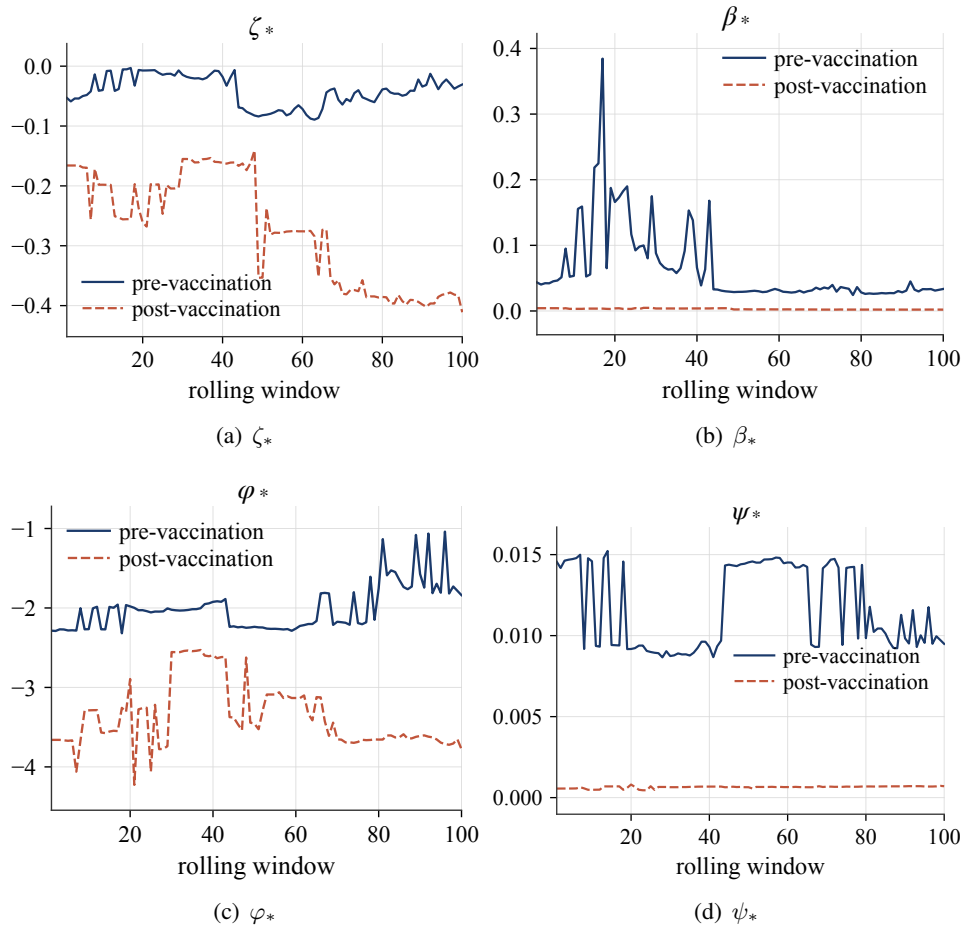


Figure 3: PARAMETER ESTIMATES OF ζ_* , β_* , φ_* AND ψ_* USING THE ROLLING WINDOW METHOD. The window length is 266 days. Each panel overlays the pre- and post-vaccination periods.

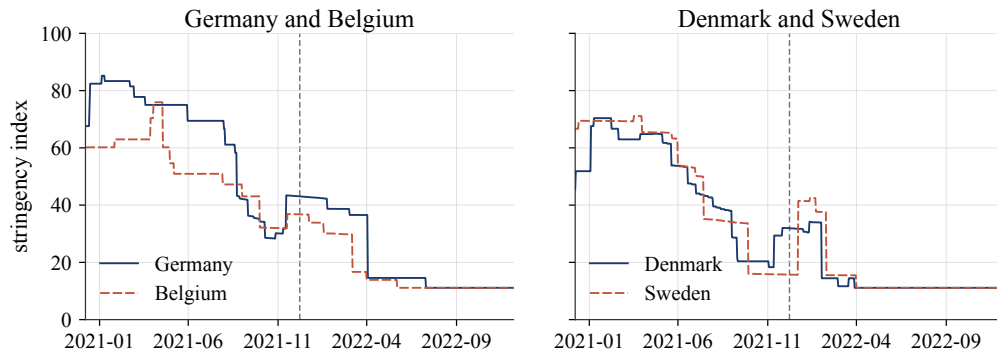


Figure 4: STRINGENCY INDEX FOR THE TWO SETS OF EUROPEAN COUNTRIES. Higher values indicate stricter measures. The data is sourced from <https://www.bsg.ox.ac.uk/research/covid-19-government-response-tracker>.

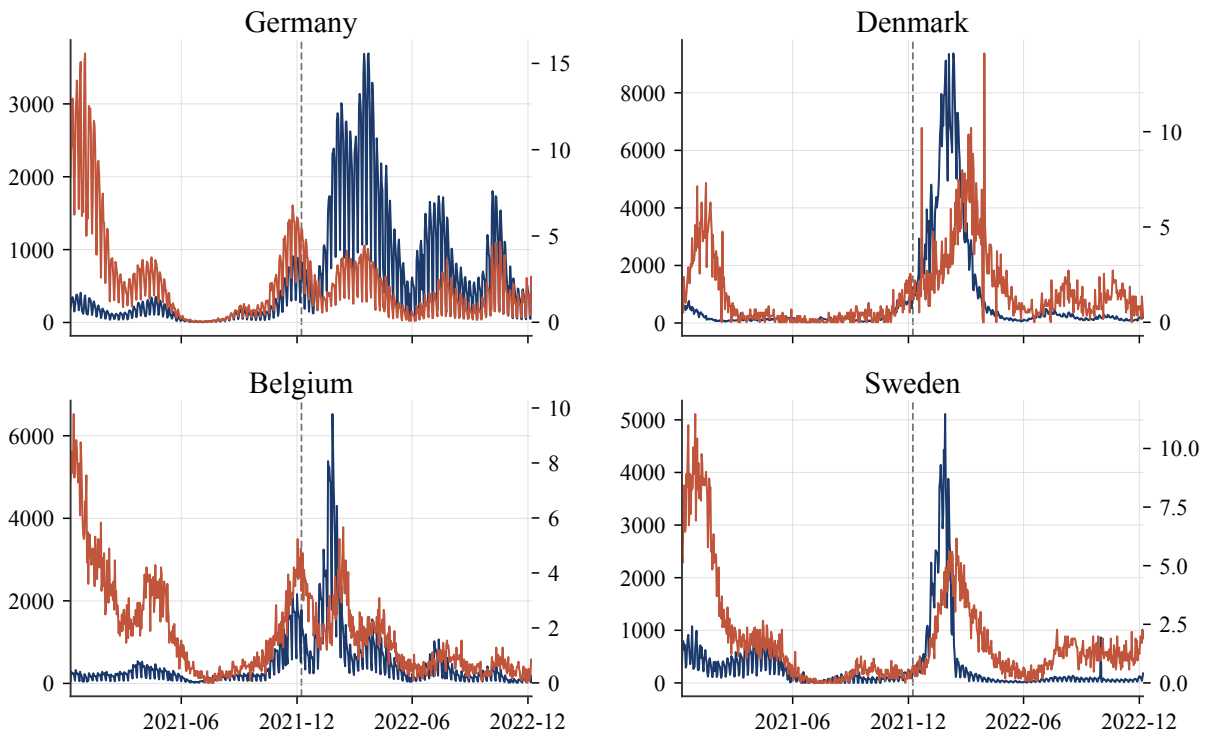


Figure 5: DAILY COVID-19 CASES AND DEATHS PER MILLION FOR THE FOUR COUNTRIES. Notes: (i) Germany and Belgium are on the left; Denmark and Sweden are on the right. (ii) The sample ranges from December 8, 2020 to December 7, 2022. (iii) The left axis measures new cases per million in red, and the right axis measures new deaths per million in blue. (iv) The dashed vertical line at December 8, 2021 separates the pre- and post-vaccination periods. (v) Source: Our World in Data; see Section A.5.1 in the Online Supplement.

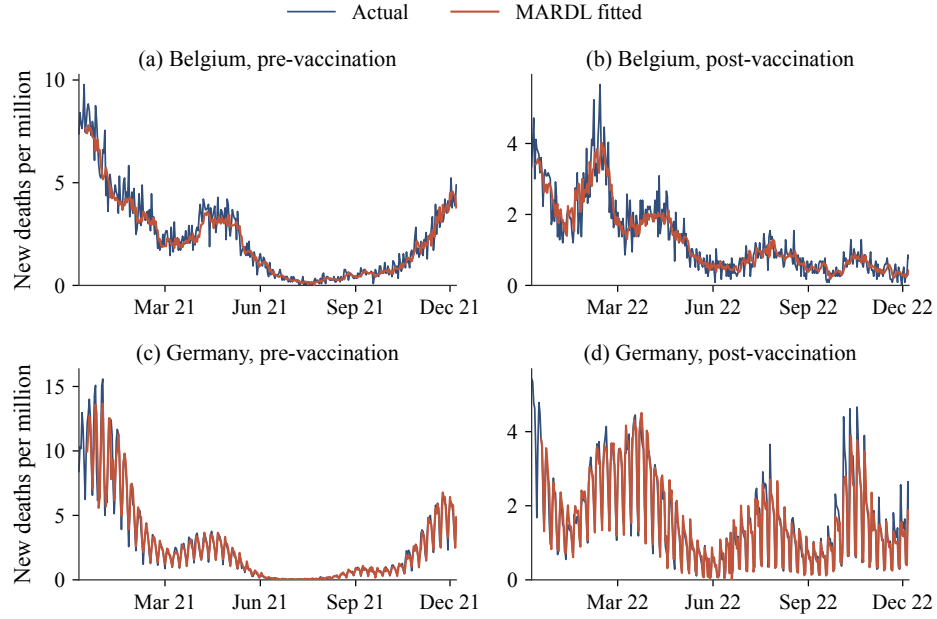


Figure 6: ACTUAL AND MARDL FITTED DAILY NEW DEATHS PER MILLION: BELGIUM AND GERMANY. Notes: (i) Panels (a) and (c) cover the pre-vaccination period and panels (b) and (d) the post-vaccination period. (ii) The vertical scale differs across panels. (iii) Conventions, periods, lag orders and data sources are the same as given in Figure 2.

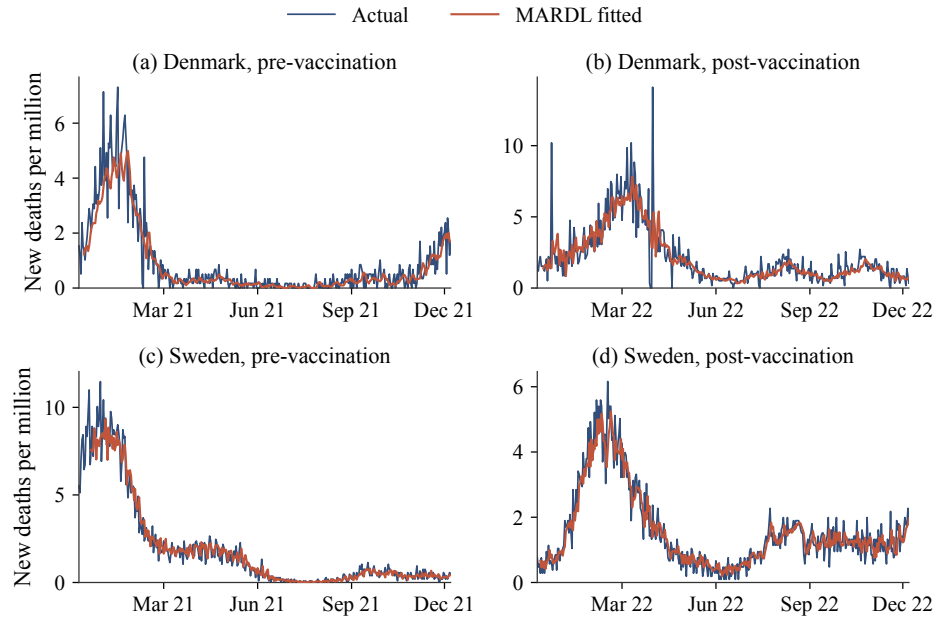


Figure 7: ACTUAL AND MARDL FITTED DAILY NEW DEATHS PER MILLION: DENMARK AND SWEDEN. Notes: (i) Panels (a) and (c) cover the pre-vaccination period and panels (b) and (d) the post-vaccination period. (ii) The vertical scale differs across panels, and negative fitted values are truncated at zero. (iii) Conventions, periods, lag orders and data sources are the same as given in Figure 2.